PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

GAMIFIED LEARNING PLATFORM FOR RURAL EDUCATION

A PROJECT REPORT

Submitted by

Mr. THEJAS MAYUR S - 20221CSE0344

Mr. MALLIKARJUN A S - 20221CSE0382

Mr. VINAY S - 20221CSE0345

Under the guidance of,

Ms. Fathima Shana E

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BONAFIDE CERTIFICATE

Certified that this report "GAMIFIED LEARNING PLATFORM FOR RURAL EDUCATION" is a bonafide work of "THEJAS MAYUR S (20221CSE0344), MALLIKARJUN A S (20221CSE0382), VINAY S (20221CSE0345)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING, during 2025-26.

Ms. Fathima Shana E
Project Guide
PSCS
Presidency University

Dr. Jayavadivel Ravi
Program Project Coordinator
PSCS
Presidency University

Dr. Sampath A K
School Project Coordinators
PSCS
Presidency University

Dr. Asif Mohamed H B
Head of the Department
PSCS
Presidency University

Dr. Duraipandian N
Dean
PSCS & PSIS
Presidency University

Name and Signature of the Examiners

1)

2)

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING at Presidency University, Bengaluru, named Thejas Mayur S, Mallikarjun A S, Vinay S, hereby declare that the project work titled **"GAMIFIED LEARNING PLATFORM FOR RURAL EDUCATION SOFTWARE"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

THEJAS MAYUR S	20221CSE0344	
MALLIKARJUN A S	20221CSE0382	
VINAY S	20221CSE0345	

PLACE: BENGALURU

DATE:

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THEJAS MAYUR S
MALLIKARJUN A S
VINAY S

ABSTRACT

Educational barriers, including limited resource access and inadequate student-teacher interaction, often hinder student motivation and academic success. *ShikshaSetu (*Gamified Learning Platform For Rural Education) addresses these challenges through a web-based, gamified learning platform designed to increase participation and improve outcomes. Built using lightweight technologies—including a Flask backend and HTML, CSS, and JavaScript frontend—the platform is scalable and accessible across a variety of devices, ensuring it remains functional even in environments with limited technological infrastructure.

The platform enhances the educational experience by integrating gaming principles such as experience points, levels, and leaderboards, alongside a Personalized Learning Model (PLM) that dynamically adjusts question difficulty based on individual performance. With real-time feedback mechanisms and a language-neutral design, ShikshaSetu supports diverse learners and encourages continuous progress. Ultimately, the project seeks to bridge the gap between traditional instruction and technology-driven education, fostering a more inclusive and effective learning environment.

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ABBREVIATIONS

CDN	Content Delivery Network
HTTPS	Hypertext Transfer Protocol Secure
IoT	Internet of Things
IoTWF RM	Internet of Things World Forum Reference Model
JSON	JavaScript Object Notation
REST	Representational State Transfer
SDG	Sustainable Development Goal
STEM	Science, Technology, Engineering, and Mathematics
UI	User Interface
UML	Unified Modelling Language
UX	User Experience
VIOLA	Virtual Intelligent Online Learning Assistant

Chapter 1
INTRODUCTION

1.1 Background

Over the last ten years, advances in digital technologies have transformed education while also enabling richer, more customized experiences. However, there remain many areas worldwide (particularly in rural areas) where schools do not use technology to communicate with their students; and therefore, those students do not benefit from such tools, including interactive and individualized teaching materials. Teaching continues to take place primarily through traditional methods in rural areas, resulting in boredom among students and a relative lack of motivation to learn. As a consequence, many students fail to reach their fullest potential [5].

As a solution to this issue, today's educational systems are rapidly adding gamification into non-gaming settings by including game-related features like points, rank, and competition in rating systems. Studies show that gamification techniques help increase student engagement, participation and knowledge retention.

Although there are many digital learning platforms out there, a lot of them don't have things like instant feedback, being able to use in multiple languages, or changing the difficulty based on the learner's level etc. This makes it hard to provide education that is both inclusive and effective, especially for students who come from different language and socio-economic backgrounds.

The system, known as ShikshaSetu, is a web based gamified learning tool developed to address these limitations. ShikshaSetu improves student engagement by providing interactive quizzes, real-time tracking of student performance and adjustable levels of difficulty. The website is designed to enhance the educational experience of students using modern web technologies and principles of gamification [1], [2].

1.2 Statistics

The need for developing innovative educational platforms is supported by the following trends or observations:

- Many learners lose interest in traditional forms of education due to low levels of interaction and engagement.
- Studies have demonstrated gamified systems can increase the participation rate of the learner while improving academic outcomes compared to other traditional/normal forms of delivery.
- The lack of access to high-quality educational resources in developing countries has created an urgent demand for low-cost, scalable solutions.
- The need for multilingual education platforms is growing in countries such as India, where students come from many different language groups.

These factors point to the importance of creating an interactive, scalable and universally accessible education platform such as ShikshaSetu [3].

1.3 Prior Existing Technologies

Numerous technology-based learning platforms and systems have been created for bettering education through technology, including:

- E-Learning Systems: They provide users with information and online courses and are often not interactive[5].
- Learning Management Systems (LMS): Systems used to deliver and assess user-generated content (UGC) that may not be as interactive as they could be for learners.

Gamified Apps: Some gamified applications and systems offer users points or rewards for completing tasks, but they do not typically have a consistent framework or immediate flexibility.

While these technologies help enhance digital education, they have some limitations including:

- Limited ability to provide personalization and adaptive learning
- No real-time feedback or analytics available
- Limited ability to support multiple languages
- No integration of features.

These limitations highlight the importance of developing one unified system that incorporates elements of gamification, accessibility, and performance tracking.

1.4 Proposed Approach

Aim of the Project

This project aims to create an interactive educational platform that will generate high levels of student engagement with improved learning results through interactive adaptive educational tools.

Motivation

The traditional education system does not provide sufficient stimulation for students when learning in a low-resource community; however, with more and more digital tools being implemented, we now have an opportunity to implement more engaging and successful learning experiences.

Proposed approach

ShikshaSetu has the following components when designing the system:

- Gamification, such as points, levels, and leaderboards.
- Ability to provide feedback and track performance in real time [7].
- Adaptive difficulty based on user's performance.
- Support for multilingual use of the system.
- Based on web technologies utilizing lightweight architecture.

The following applications will be developed throughout the project:

- Digital classrooms/e-learning;
- Rural and remote education systems;
- Skill development/training programs;
- Educational institutions utilizing ShikshaSetu to promote interactive learning.

The limitation of ShikshaSetu includes:

- Requires access to the internet to achieve all functionality;
- Requires development of multilingual content in order for the system to be fully functional; and

1.5 Objectives

Behavior:

Create an engaging interactive learning system with a focus on gamifying study material to encourage learners' motivation.

Analysis:

Use real-time data to allow for user performance evaluation and provide improvements.

System Management:

Setup a central location to manage user information, quizzes, and performance data.

Security:

Establish secure methods of authenticating users and handling their data.

Deployment:

Implement a scalable web-based solution, users can access the system from many different types of devices.

1.6 SDGs

Fig 1.1 Sustainable development goals

SDG 4: Quality Education

SDG 4 aims to provide equitable access to quality education for all individuals and encourage lifelong learning. By changing traditional pedagogies into a more interactive and fun way of learning, the proposed system will help reach that goal [5].

ShikshaSetu enables students to interactively learn through gamification. Students will be rewarded for their participation in quizzes, challenges, and performing tasks. On this site, students can engage in continuous learning through the levels, rewards, and feedback mechanisms. Consequently, they will have improved retention of information and be more motivated to learn.

The platform allows students to learn in their preferred languages. By breaking down the language barriers typically associated with learning, more students will be able to take advantage of the learning opportunities provided. This will have an even greater impact for students in underserved, rural areas of the country.

Furthermore, students can use the platform to learn at their own pace and based on their individual strengths and weaknesses. By providing students with this personalized learning experience, there will be a better understanding of the learning concepts and knowledge gaps will be closed.

SDG 9: Industry, Innovation, and Infrastructure

SDG 9 targets the growth and expansion of durable infrastructure, encouraging productive and inclusive industrialisation, along with fostering creativity/innovation through new and flexible platforms. In this context, ShikshaSetu aims to utilise modern technology to develop a highly scalable and efficient educational platform.

The design of the system architecture is through a standard web-based structure and use technologies typically associated with web page authoring such as Flask (web application framework), HTML, CSS, and Javascript, which provides a lightweight and efficient execution model. The technology used to deliver the education is based on real-time data analytics, adaptive learning methods, and a modular gaming model. Therefore, the ShikshaSetu Platform demonstrates a high level of innovation in the educational technology field.

Additionally, the platform is built on a highly scalable technology platform that allows for extremely high user capacity and thus can be used in all educational settings including schools, universities and vocational/training centres.

Also, utilising data-based intelligence enables the platform to assess student performance and create improved learning methodologies thus leading to improved quality educational infrastructure [3].

SDG 10: Reduced Inequalities

SDG 10 aims to reduce inequality both within countries and across the world. The platform will help achieve this goal by enabling equal access to educational materials to all students, regardless of their backgrounds.

ShikshaSetu is designed to work well in low-resource environments which are typically found in rural or economically disadvantaged areas. With its small file size, the platform will continue to function well when there is limited access to the internet.

Multilingual support ensures that students are not limited by language barriers and that they have the same opportunities to learn regardless of their location, economic status, or educational level.

By making quality education more accessible and affordable, the platform will help to close the gap between urban and rural education systems.

SDG 11: Sustainable Cities and Communities

Sustainable Development Goal 11 calls for cities and communities to be safe, secure, resilient and sustainable. Education is a driver of sustainable communities, and ShikshaSetu is part of promoting digital learning practices in support of this goal.

The platform supports an environmentally sustainable education through reducing reliance on the use of printed materials such as textbooks, as well as promoting digital literacy for the creation of modern smart cities.

The Platform can be used in schools and community learning centers to help develop a digitally empowered community.

1.7 Overview of Project Report

Chapter 1 - Introduction

In this chapter, the author explains all aspects of a project, including background information, motivation and reasons for creating a game-based educational system. The author describes what issues are being solved, lists goals for accomplishing project objectives, and describes how the project supports Sustainable Development Goals (SDG). Finally, it provides a general overview of how the entire report will be structured.

Chapter 2 - Literature Review

This chapter reviews past literature and published materials about gamification used in education, digital learning platforms, and interactive systems. It discusses the methodologies used in previous studies, contrasts the various approaches to gamification, and outlines the limitations and gaps in research that created a foundation of knowledge for the development of the system proposed.

Chapter 3 - Methodology and System Design

This chapter describes the development methodology for the project. It outlines the general development process for the project, including requirement gathering, system design, implementation plan, and testing activities. The model used for development provides for a systematic approach to developing the project and provides for good project management.

Chapter 4 - Implementation and Development

This chapter discusses the planning and execution aspects of the project. It describes the project schedule, task assignments, risk analysis, and budget estimates. Efficient use of all resources is discussed in order to achieve effective completion of the project.

Chapter 5 - Results, Testing, and Evaluation

This chapter contains an in-depth analysis of the requirements of a proposed solution as well as the design of the solution itself. Included in this analysis are details of the system architecture, block diagrams, data flow, and functional components of the proposed system. The design decisions made will be reviewed in terms of performance, scalability and usability.

Chapter 6 - Conclusion and Future Enhancements

This chapter will provide details on the tools, technologies and platforms used to create the project. This includes an explanation of both frontend and backend implementation, integration with databases, and functionality. This chapter also will provide insight into the development and integration of each individual module within the project.

Chapter 2

LITERATURE REVIEW

2.1 Gamified Learning Systems

Some studies in this direction explore in-depth the ways gamification can enhance education. They look at different gamification elements such as points levels badges, and leaderboards and scenarios where these are most beneficial. Generally, they advocate making learning more stimulating and engaging through gamification as opposed to conventional methods. According to the researches, students trust gamified environments and show increased enthusiasm and active involvement. Rewards trigger the desire to learn endlessly and also contribute to efficient memorization.

On the other hand, quite a few of these experiments are mainly adding game features and totally disregard the introduction of adaptive learning technologies. Consequently, the educational journey wouldn't be customized for individual students which in turn will make the learning less effective especially for those with different skills [1],[2].

2.2 Traditional E-learning Platforms

Well-organized e-learning systems are the most common method for delivering educational content via digital platforms. These platforms offer organized learning resources such as videos, notes, and quizzes.

While they have the benefit of providing education access irrespective of location, they can be quite devoid of interactivity and engagement. Usually, these platforms deliver the same material to everyone without any adaptations based on the individual's performance. The lack of instant feedback and gamification aspects has indeed placed them at a disadvantage when it comes to sustaining learner's interest and motivation over a long period of time [5].

2.3 Gamification with Leaderboard and Rewards

Besides, the research has also looked into the systems implementing leaderboards and rewards to increase user activity. These systems set up a competitive setting where users are encouraged to raise their performance to get better standings [3].

Studies show that these characteristics not only raise the level of involvement but also motivate the users to finish the learning tasks. Nevertheless, these systems tend to emphasize more on

the competition rather than the educational results. Also, they are not always delivering the relevant feedback or modifying the content, which decreases their effectiveness for education.

2.4 Web-based Learning Systems

Lightweight web-based learning platforms have been created to ensure that learners can be reached through multiple devices. These platforms employ simple software architectures that can perform well even in less resourceful settings. Hence, they can be used even in rural areas where there is a lack of proper facilities.

These platforms definitely can increase the reach of education but still, they may not be able to handle large numbers of learners sustainably. More users accessing the platform at the same time will lead to scaling and performance problems. Besides, some platforms may not be equipped with hi-tech features like real-time analytics and learning personalization [4].

2.5 Multilingual Educational Platforms

Nowadays, multilingual support has become a very effective feature in the educational area of learning systems. Besides, research indicates that making content available in more than one language is a great way to enhance accessibility and give people of different cultures and ethnicities a chance to understand the material better.

Such systems are capable of lowering the language barrier, which is one of the obstacles to inclusiveness, and at the same time, they encourage inclusive education. Nevertheless, the development of such support involves not only accurate translations but also the regular updating and renovation of the language files. The majority of the current systems feature only a few language choices, and thus, their use is limited to a small number of people, excluding a wider audience [6].

2.6 Real-time Performance Tracking Systems

Some research has been carried out on the incorporation of real-time analytics into learning platforms. Such platforms keep a record of user activities like quiz performance, timing, and accuracy. The data gathered is utilized to give feedback and enhance learning results [8].

Real-time monitoring allows students to immediately identify their strong and weak points. Besides, it gives teachers the opportunity to continuously keep track of students' learning. On

the other hand, the integration of analytics with other parts of the system is very complicated and certain systems have difficulties in providing efficient real-time updates.

2.7 Modular System Architecture

Modular architecture is extensively employed in software development these days. In the context of educational software, it enables various parts like the frontend, backend, and database to work separately. This leads to higher system maintainability as well as scalability.

Studies suggest that modular frameworks are simpler for updating and upgrading. On the other hand, if the integration of modules is not handled correctly, it might cause communication problems and a less efficient system. Hence, thoughtful design and synchronization become necessary [10].

2.8 Adaptive Learning Systems

Adaptive learning systems modify the challenge level of the material according to how well the user is doing. Besides individualizing the learning experience, these systems also allow users to move forward at a pace that suits them one.

According to research, adaptive systems enhance learning effectiveness and increase user satisfaction. On the other hand, the creation of these systems entails sophisticated algorithms and thorough data analytics, so this is why a large number of the existing platforms do not implement adaptive learning features completely [7].

2.9 Interactive Quiz-based Systems

Quiz-based learning systems are commonly used for knowledge assessment and enhancing learner engagement. They offer immediate feedback and enable users to engage in multiple practice sessions.

The dialogue form of quizzes is instrumental in explaining the materiel and helping learners remember better. Yet, some systems do not integrate well other features such as analytics through gamification, which ultimately reduces their effectiveness [8].

2.10 Integrated Learning Platforms

Updated studies show the demand for unified solutions that feature gamification, multilingual support, analytics, and adaptive learning integrated into one system. Such systems are targeted at delivering a holistic learning solution.

While some steps have been taken, a lot of present platforms still emphasize individual features. This leads to missing out on a thorough and potent learning experience delivery [3].

Summary of Literatures reviewed

SI	Article Title & Year	Method	Key Features	Limitations	Project Implications
1	[1] Beronova (2025)	Offline Gamified App	Improves STEM engagement, offline learning	No analytics, no multilingual	Web-based platform, analytics, multilingual support
2	[2] You & Sanmugam (2024)	Gamified Learning	Improves motivation, rural focus	No system architecture, no tracking	Full platform with dashboards & analytics
3	[3] Santos et al. (2024)	Personalized Gamification	User-based game experience	No rural/offline support	Adaptive learning & offline modules
4	[4] Capecchi et al. (2024)	Serious Game	Improves critical thinking	Not STEM-focused	Gamified STEM modules & UI design
5	[5] Palee (2024)	STEAM Gamification	Improves creativity & engagement	No analytics, unclear system	Performance tracking & structured modules
6	[6] Sales & Oliveira (2025)	Literature Mapping	Improves engagement & learning	No standard system, weak evaluation	Standardized platform & evaluation system
7	[7] Papadakis (2022)	Systematic Review	Improves motivation in STEM	No real implementation	Full gamified STEM platform with analytics

Table 2.1 Summary of Literature reviews

Chapter 3

METHODOLOGY

Building the ShikshaSetu system is a planned and thorough process which is followed attentively to make sure that all parts are well-designed, implemented, and tested. Since the project is composed of various tightly linked modules including Interactive Learning System help, central resource managing, user verification, and cloud deployment, it is very important that a method be chosen which maintains transparency, uniformity, and excellence at all times during the process of development. A proper method serves not only to arrange the sequence of work but also makes sure that each phase of development is checked before the next stage is started. This leads to the V-Model (Verification and Validation Model) was picked as the development methodology. It is a very clear model showing how each phase of development is related to the corresponding phase of testing. In this way, every part is verified and validated in an orderly fashion.

Selection of V-Model Methodology

Step 1: Understanding the Nature of the Project

The ShikshaSetu system consists of a number of inter-related modules including:

- Each of these parts need detailed planning,
- well-defined requirement, and regular testing to confirm seamless integration.

As the whole system is a mix of user interaction and backend processing, it would be best to test each module very carefully.

Step 2: Reason for Selecting V-Model

The reasons for selecting the V-Model are:

- It offers a systematic and rigorous way to develop a system
- Every development step is associated with a specific testing phase
- Experience mistakes are able to be detected and fixed at initial phases
- It guarantees the system is of very high quality and reliability
- It fits perfectly for academic projects especially when the requirements are well known at the start

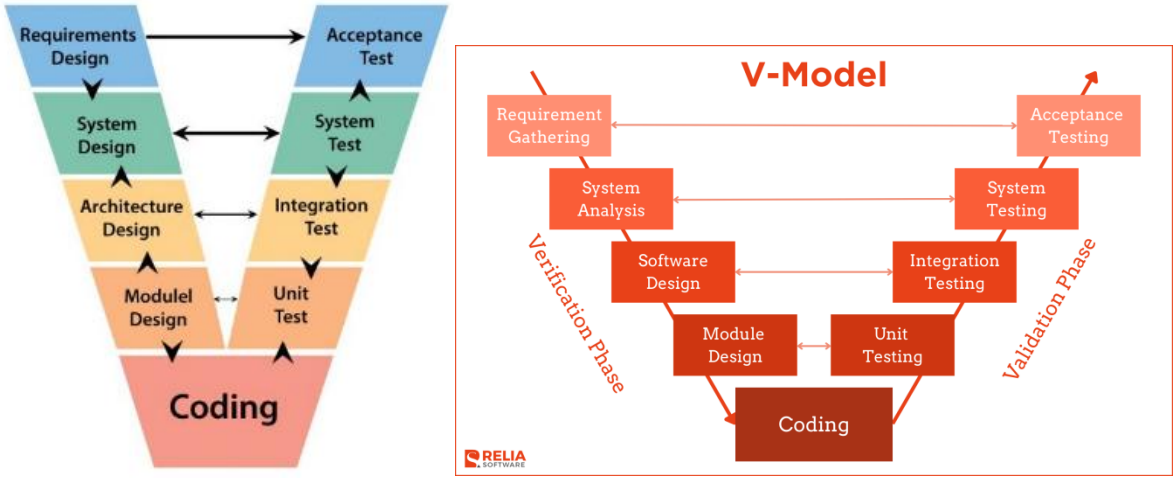

Fig 3.1 The V model methodology

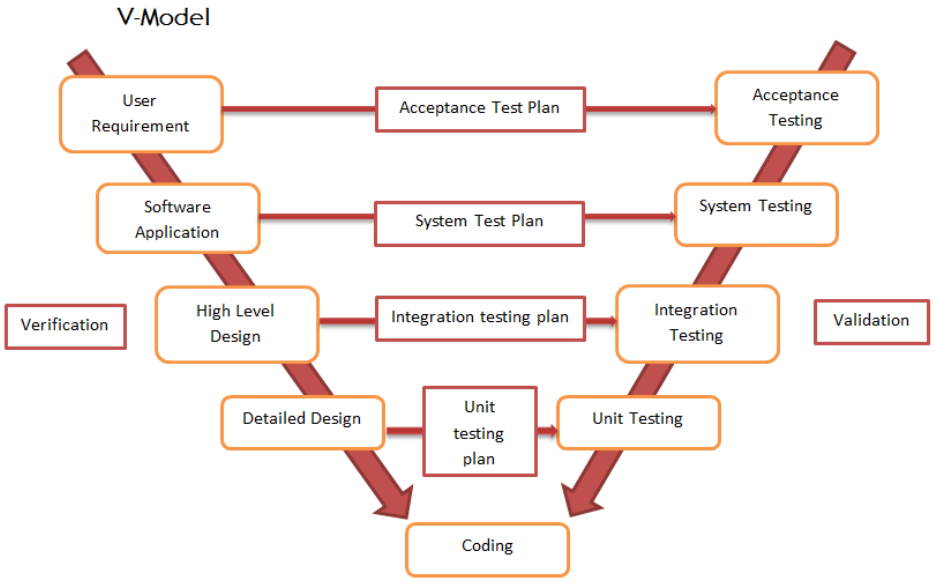

Fig 3.2 Another example of the V model methodology

The V-Model consists of two main phases:

- **Verification Phase (Left Side)** – Focuses on system design and development
- **Validation Phase (Right Side)** – Focuses on testing and verification

Each stage on the left corresponds to a testing phase on the right, ensuring that every component is validated properly.

Verification Phase (Left Side of V-Model)

Step 1: Requirement Analysis

Requirements are finalized first, setting the foundation for everything else. Teams gather facts about user actions and system goals during early planning, laying out exactly what needs to work and how people will engage with it on the ground.

Key activities:

- Functional needs are set after seeing how students, faculty.
- Admins actually use the platform, and Interactive Learning System tasks like answering questions and helping users find info come from real usage patterns.
- Features including accessing materials, getting alerts, and logging activity emerge during planning.

Result:

The system starts with a clear plan built around exact needs, not guesswork.

Step 2: System Design

This plan shapes how everything fits together - what parts do what, and where they connect.

Key points:

- Architecture sets up front-end, back-end, and database layers
- Layout and menus follow user behavior patterns
- Interactive Learning System steps move naturally from start to finish
- Tools like Flask, PostgreSQL, HTML, CSS, JavaScript are picked for fit and speed
- Data moves between pieces using consistent paths

Result:

The big picture comes alive through structured component relationships.

Step 3: Architectural Design

Dividing the system into modules like Interactive Learning System, UI, database, and authentication ends up shaping how everything connects together.

Key activities:

- Splitting functions into separate parts for easier handling Planning
- API calls so frontend talks directly to backend
- Graphing out where data flows across systems
- Maps out which pieces rely on each other

Result:

An architecture blueprint that clearly shows component links and communication paths [10].

Step 4: Unit/Module Design

In this step, each module is designed in detail.

Key activities:

- Interactive Learning System logic built around intents, response flows, and fallback paths
- Databases laid out with users, resources, and log entries
- APIs set up for specific actions and data access
- UI elements designed with how users will actually engage them
- Data checks and errors handled upfront

Result:

Clear internal design of each module so coding can begin smoothly.

Validation Phase (Right Side of V-Model)

Step 5: Unit Testing

Every module undergoes testing separately.

Examples:

- Test Interactive Learning System intent recognition for different languages.
- Testing user authentication system
- Verify database CRUD operations.

- Checking UI responsiveness

Result:

Each unit is functioning well independently.

Step 6: Integration Testing

Once unit testing is done, the different modules are merged and tested together.

Examples:

- Interactive Learning System ↔ Backend integration
- Frontend ↔ Database communication
- Authentication ↔ User data management

Result:

Modules work together without errors [8].

Step 7: System Testing

The system in its entirety is subjected to testing at this level.

The key tests are:

- Navigating a completely multilingual website
- Database performance with multiple requests
- Security testing (login, encryption, session handling)
- Performance testing under load

Result:

It is the very last test that confirms that the entire system is working properly.

Step 8: User Acceptance Testing (UAT)

Real users test the system to validate its usability and effectiveness.

Activities:

- Interactive Learning System features get used daily by students

- Usefulness is measured through real feedback
- System updates follow user insights

Outcome:

Last consent and recommendations for the release stage

Summary:

Testing begins before any code is written, locking in requirements early. The V-Model runs design and validation side by side, catching issues fast. Each module gets checked twice - once in plan, once in practice. Outcomes stay stable even under pressure, and in the end, users get something that works and grows easily.

Chapter 4

PROJECT MANAGEMENT

4.1 Project Timeline

ShikshaSetu's Interactive Learning System functionality, centralized academic resource management, user authentication, database integration, and cloud deployment all needed tight alignment during planning. That said, getting those parts working together wasn't just about picking tools - it required mapping out how each piece fit into the overall workflow. The team broke down the process into clear steps: requirement gathering, literature survey, possibility analysis, system architecture design, tech stack selection, database schema planning, and user interface design. As it happened, this structure helped avoid delays caused by unclear expectations.

Every module got attention without overlapping efforts that could confuse timelines. Still, some parts ended up taking longer than expected because dependencies weren't fully visible early on. Planning made the work more organized and reduced chances of missed deadlines. End results were cleaner when tasks had defined start and end points.

Progress is tracked more clearly when tasks appear side-by-side on a gantt chart. The chart shows how long each step takes, where they line up, and when overlaps happen. Dependencies between steps come out plainly, so key work doesn't get skipped or delayed. Resources end up assigned smarter because timing is visible and shared across teams.

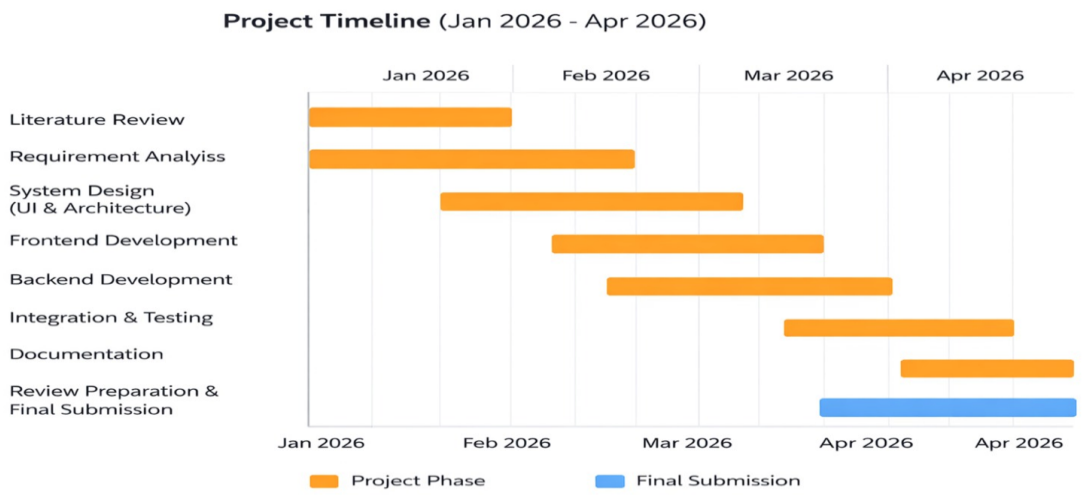

Table 4.1 Project planning timeline

What actually ended up happening was that the ShikshaSetu planning skipped formal milestones, just moved forward based on gut instinct and urgency. Although timelines stretched across weeks, tasks didn't follow strict order, and instead, they evolved with feedback from real users. As it happened, the first move wasn't a checklist, it started with talking directly to students, faculty, and admins about daily struggles like broken communication channels and missing digital tools. The team saw gaps in current academic setups: scattered data, no live help, and no single hub for learning content.

Still, the design stayed flexible despite early plans. Since decisions changed often after discussions, new features popped up without official approvals. Each update built on what worked best in actual use - not theory. It wasn't a polished roadmap at all - more like a living document shaped by reality[10].

But most of the early work wasn't about building - just understanding what already existed. The team dove into existing tools like Learning Management Systems, Interactive Learning System-driven setups, and cloud-based education apps to see how they worked, what they offered, and where they broke down. Realistically, the biggest flaw, while a total disconnect between communication flows, resource handling, and smart support features. That gap didn't just show up on paper, it became the foundation for how ShikshaSetu would operate from day one.

Things fell apart during early testing because the tech stack couldn't handle peak loads. That happened after the team picked hardware too slow for daily use and didn't test stress scenarios. The budget ran dry when they underestimated maintenance costs. This turns out, even good designs fail if the numbers don't match up. At least in theory, balancing cost and function works - just not here, and with real-world data, decisions changed fast.

In this stage, the overall structure of the system was defined, including how frontend, backend, and database parts would work together. The architecture design laid out exactly where data moves across modules How incoming user requests get handled. Once practicality was confirmed, development kicked off with a clear path forward. For better or worse, the plan prioritized scalability and ease of maintenance. Overall, the setup balanced efficiency with long-term upkeep. With consistent flow mapping, every component had its place in operation.

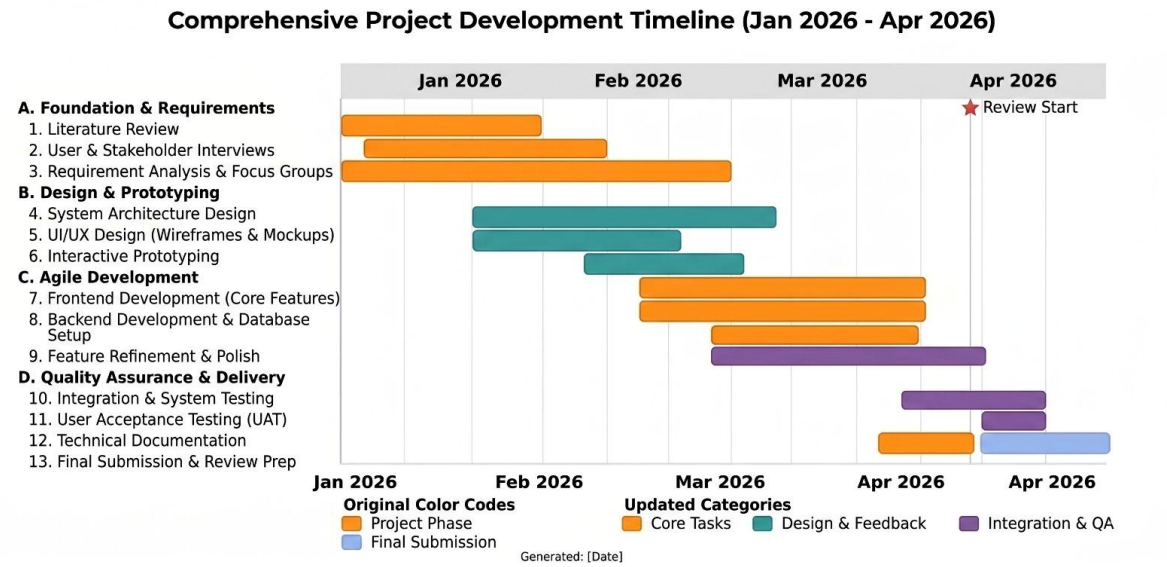

Table 4.2 Project implementation timeline

Flask (Python) was picked for the backend because it's light and easy to shape. PostgreSQL stayed as the database due to how well it handles organized data over time. HTML, CSS, and JavaScript formed the front end to keep things smooth and accessible across devices. With this mix, performance and growth potential stayed strong throughout development. In practice, decisions were made through testing rather than theory. The stack worked out in real life without breaking down under load. Mostly, changes didn't slow progress much, and it all held up even when usage spiked suddenly.

After that, the Database Schema Design phase was carried out. At this point we decided how the database should be organized for storing user details, educational materials, Interactive Learning System info, and system logs. Various tables were set up along with correct relationships among them to make data retrieval fast and to maintain data integrity. This stage was very important for making sure that the system would be able to process data properly and also be expandable.

UI/UX Design is the last step in the planning phase, which is all about designing a user interface that is both intuitive and pleasant to the eye. During this phase, the team produced wireframes and layouts to design the user journey and interaction with the system. Achieving simplicity, responsiveness, and ease of navigation on various devices like desktops, tablets, and smartphones was the main aim. A good user interface design not only creates a better user experience but also makes sure the system is accessible to a broader audience.

The Gantt charts depicting the schedule of the project clearly reflects the order and time of each task. Usually, the schedule covers from Week 1 to Week 7, with the tasks placed in the order of logic. The activities start with collecting requirements, then conducting a literature review as well as a feasibility study. After the initial stages have been finalized, the project goes on to system architecture designing, technology selection, and database planning. Planning is wrapped up with UI/UX designing, a step that gears the system up for the implementation phase.

However, it should be stressed that few of the activities mentioned in the timeline overlap intentionally for the purpose of achieving much higher efficiency, e.g. technology stack selection was paired with system architecture design while database planning coincided with UI/UX design. Besides that parallel execution of tasks allows reduction of overall time without sacrificing quality of work. The timeline thoughtfully laid out ensured not only systematic implementation of each phase, but also paved the way for the development process clearly. Hence, by sticking to this plan, the team was capable of keeping the same style, monitoring progress very well and making sure that all project targets were accomplished within the deadline.

4.2 Risk Analysis

Factor	Impact on ShikshhaSetu
Political	Supportive government policies for digital education and rural development.
Economic	Budget constraints for development and deployment.
Social	High need for engaging and inclusive learning platforms.
Technological	Rapid advancements require scalable and up-to-date solutions.
Legal	Data privacy and security regulations must be followed.
Environmental	

Table 4.3 Example of PESTEL analysis

The created PESTLE figure pretty much shows the six external risk categories divided into separate colored columns. The sections are indicated by a capital letter (P, E, S, T, E, L) in bold, which makes it easy to recognize the structure at a glance. The first thing under each heading is the identification of the potential risks and the corresponding mitigation measures in bullet points from your project.

The composition employs bright colors like those in the reference image you sent, thus each factor is quite distinguishable though the layout is still neat and of a professional nature. This way of presenting the PESTLE analysis not only makes it easier to understand but also contributes to the clarity of your report by showing the complex information in a well-organized, reader-friendly format [4].

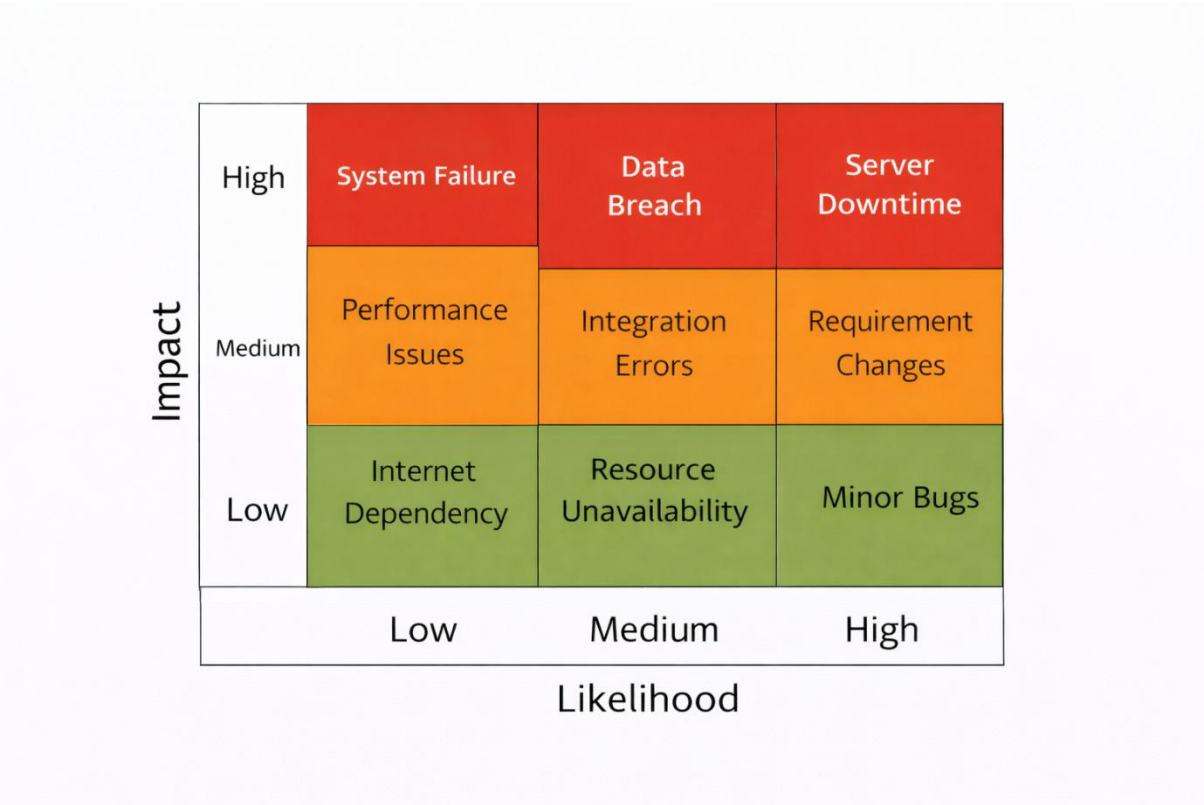

Table 4.4 Example of Project phase risk matrix

Integration issues between modules can trip up the system when components don't talk well. Bugs appear during development, and performance dips under pressure - both happen in practice.

Device compatibility fails if layouts break on smaller screens. A bad interface hurts how users engage with features.

Interactive Learning Systems often stumble on unusual questions or deep topics because their rules don't stretch far enough. The issue comes down to rigid programming without real-world flexibility - fixes include smarter logic and ready-made answers when things go off track. Security flaws pop up when user info gets exposed through weak locks or broken backups. Usually, data stays safe thanks to strong encryption, strict login checks, and careful database oversight. Service crashes can happen if servers fail or cloud providers underperform. Relying on outside hosts creates risk - this is managed by picking stable providers and having backup plans in place.

4.3 Project Budget

The budget ended up being tight, so made sense given how much reliance the ShikshaSetu system placed on open-source tools. Costs stayed low because software development formed the backbone of the project. Cutting down on licensing fees and expensive licenses. With internet access and basic hardware kept minimal, spending didn't spike despite growing needs. That said, every item, from tool subscriptions to document storage - was tracked closely so nothing slipped through. The plan felt solid once all major expenses were laid out in detail. Even if some small gaps remained at first.

Step 1: Identification of Tasks and Resources

Resources were mapped directly to every stage of system building. Frontend work, backend code, Interactive Learning System logic, so database links, testing phases, docs, and rollout needed careful planning. Tools, devices, internet access, and printables like paper manuals made up the list. In practice, software and online systems covered nearly all costs. Most spending went toward digital setups for the tech-heavy setup.

Step 2: Check Team Availability

The project didn't need extra spending on gear. Team members already had laptops and could tap into the dev tools they needed. Tools like Flask, PostgreSQL, and HTML, CSS, JavaScript were open-source and free to use. Cloud hosting offers free plans too, which cut down costs right away. Generally, everything was accessible without paying upfront. That kept the budget tight without sacrificing functionality. Sometimes people still worry about licensing, but in practice, it's not a concern here. Resources worked out well without any delays.

Step 3: Estimate Task Duration

Stable connectivity was needed for testing and deployment. The project runs over several weeks, so internet use stayed a fixed expense. Continuous access helped keep operations running smoothly. Other tasks relied on consistent online performance. All this shaped how the total budget came together.

Step 4: Use Experience and Previous Data

Without clear guidelines, early estimates drifted off course. Through borrowed designs and known spending habits, the team built a solid picture of expected costs. Open-source platforms and free cloud access cut out any hidden bills entirely. Ultimately, everything stayed low - just enough to cover daily tasks without big pushes.

Step 5: Project Budget Estimation

Documentation and printing costs made up a big chunk of the budget, with internet and data usage across the entire project span. Expenses for miscellaneous items were added in as well. Hardware and software tools were already in place and free, so those didn't need to be factored in.

Project Budget

Category	Resource / Tool	Cost	Remarks	Mitigation
Software Tools	Flask, PostgreSQL, Vercel, Render	₹0 dium	Moderate	Open-sourc / Free-tier
Development Devices	Laptops (Student-owned)	₹0	Moderate	No additional purchase
Internet * & Utilities	Documentation, printing	₹1,000	High	Project submissions
Total Estimated	**Cost**	**₹1,500**	Low-cost academic project	

Table 4.5 Example of project budget

The picture showcases a thoroughly organized Project Budget Table, which is intended to look like a professional budgeting template. The arrangement features columns for Category, Resources/Tools, Cost, and Remarks that are well separated from each other in such a way that one can easily grasp the financial distribution of the project just by glancing at it.

Every line is associated with a particular idea of how the system development costs should be broken down into software tools, devices, utilities, and miscellaneous needs. The table indicates that most of the development tools (Flask, PostgreSQL, Vercel, Render) are without any charges because they are open-source and free-tier available. As a result, the overall project budget is cut down considerably. What's more, no other expenses are provided except for internet utilities and miscellaneous documentation needs.

The very last line of the table sums up the Total Estimated Cost of ₹1,500, thus pointing out that the project is very cost-efficient and can meet the needs of the academic sphere. The neat layout and the gentle color accents make the table attractive, readable, and suitable for direct insertion into your project report or presentation.

Costs drop sharply for ShikshaSetu because the team uses open-source tools and saved resources. The plan works well without sacrificing quality or practicality. In practice, this makes funding easier and keeps delivery on track more or less.

Chapter 5

ANALYSIS AND DESIGN

1. System Analysis

System analysis sets the foundation for building effective software. Helping teams grasp pain points, collect user needs, and map out required features. It doesn't just observe the status quo, it spots gaps, analyzes inefficiencies, and designs responses that fix them. For ShikshaSetu, this means tackling how old academic tools fall short: scattered messages. No shared databases, slow updates, and zero live help. Students often spend too much time digging through outdated records or waiting days for feedback. Realistically, without a unified space where info flows instantly Users get immediate answers, progress stalls. Generally, modernizing workflows isn't optional - Mainly when education depends on timely access and clear communication.

Information gets lost because students jump between emails, apps, and notice boards instead of having one clear source. Mostly, they don't know when something's posted or how to find what they need. Slows down work and causes stress, so there's no smart tool on hand to answer basic questions right away. Students end up asking professors or staff repeatedly. A new Interactive Learning System-driven academic space solves this by letting people talk directly to it in plain language. It gathers everything under one roof so users can get help whenever they want without switching tabs. Features picked during review include live support, instant search, and resource tracking.

2. System Design

The ShikshaSetu system uses a client-server setup so users see the frontend even as the server runs the backend tasks. Components are built separately and joined together later, following a modular, scalable method. This structure lets teams work on parts without disrupting others. System design turns initial needs into a clear plan with defined architecture, data paths, and interface features. Generally, this blueprint keeps development focused and makes sure every requirement gets addressed. Data moves smoothly between parts, and user experience stays consistent across functions. In practice, independent modules help reduce errors during integration.

a. Architectural Design

PostgreSQL holds user data, academic materials, and logs in a structured format. Backend logic runs by Flask (Python), so taking incoming requests and passing them through to the database. Frontend tools like HTML, CSS, and JavaScript build the clickable interface users see daily. Cloud services handle hosting, offering reach across regions without extra setup. In practice, this setup works well usually, scalable, stable, and easy to access on demand [10].

b. Functional Design

Functional modules include:

- **Interactive Learning System Module:** Handles user queries and provides real-time assistance.
- **Resource Management Module:** Provides access to study materials and updates.
- **Notification Module:** Sends alerts and updates to users.
- **Authentication Module:** Facilitates user sign-up, login, and profile management by securely handling encrypted data.
- **Database Management Module:** Stores and retrieves system data.

c. Security Design

Often, people forget how fragile user data can be, In particular when sitting in plain sight inside a database. The setup uses strong passwords, strict login checks, and access rules based on roles to limit who sees what. Communication between devices runs over HTTPS so no one intercepts messages mid-transit. Everything saved ends up locked behind encryption, blocking hackers from peering inside.

d. Interface and Data Design

Smaller screens are handled smoothly, so logs load fast on phones too. The app works well on laptops, tablets, and mobile devices without slowing down. Data lives in tables linked by keys - users, resources, logs stay connected. Indexes keep the connections sharp so searching doesn't stall. As it happens, performance stays strong even when many records pile up.

e. Deployment Design

The system runs on cloud platforms for global access and to scale with traffic. Frontend services live on vercel, backend logic and data storage sit on render. Overall, this setup keeps things running smoothly even when users surge unexpectedly. Still, downtime remains rare due to redundant layers across zones.

3. Relationship between Analysis and Design

In the ShikshaSetu project, design decides how features are built, although analysis figures out what needs to be delivered. This turns out, both steps run hand-in-hand during software creation. Pretty much, they feed off each other from start to finish.

In this work:

- Needs like instant help, shared tools, and locked-in login were spotted early in the review.
- During build-out, those demands shaped a layered setup - clear parts, how data moves, and how people engage.

As it happens, this shift turns problems into working plans. Often the gap between spotting what's needed and turning it into code closes cleanly. That said, results end up stable and grounded in reality [4].

5.1 Requirements

System Purpose

Most students still spend hours hunting for info across scattered emails and offices - now that's changed. The ShikshaSetu system brings everything together in one place, letting learners find classes, grab notes, get alerts, and talk directly to faculty using a smart Interactive Learning System. Instead of jumping between forms or waiting for replies, everything flows through a single portal. Mostly, it cuts down admin work and keeps help available at any time without needing logins or appointments. Built with current web tools and cloud backing. The platform works smoothly whether it is on campus or halfway across the country. For better or worse, learning feels more connected than ever before.

System Behaviour

Students and faculty log in to exchange messages using the web interface or Interactive Learning System. The ShikshaSetu platform runs in two main states, live engagement and background sync.

1. Interactive Mode (User Mode)

During live engagement, real-time conversations happen between users and the system. Background sync keeps data updated without user action usually.

Behavioural Characteristics:

- User sessions get logged automatically so teams can study behavior patterns over time.
- The system tracks every click, query, and login without prompting for permission.
- Students find notes and class alerts right where they need them, no hunting through folders.
- Access is open on phones, tablets, or laptops at any moment. Progress reports appear as soon as activities happen.
- Users don't have to manually check in.
- Real-time updates keep things clear and current.
- If someone types something vague or off. The bot steps in with a gentle suggestion instead of failing silently.

2. Administrative Mode (Backend Mode)

This mode is designed for administrators and faculty members who manage the system.

Behavioural Characteristics:

- Backup and recovery mechanisms ensure data reliability and system continuity.
- User activity logs are reviewed regularly so admins can track usage patterns.
- Faculty upload study guides and send class updates directly to students.
- Response logic gets tweaked based on how Interactive Learning Systems perform over time.

- Access to student records remains restricted unless authorized by policy.

System Hardware Requirements Phase

Requirement Category	Description
Initial Conditions	Internet-enabled device (desktop, laptop, or smartphone) with a modern web browser
Input Parameters	User queries, login credentials, resource requests, Interactive Learning System interactions.
System Outcomes	Interactive Learning System responses, access to academic resources, notifications, user data retrieval.
Constraints	Stable internet connection, device compatibility, browser support

Table 5.1: System Hardware Requirements Phase

With just a smartphone or tablet online, anyone can join ShikshaSetu. No extra downloads, no special setup, just open the browser and go. For better or worse, this keeps it simple and works almost anywhere.

Initial Conditions:

These are the essential requirements that the users must have to be able to access the system. As the platform is online, it just needs a device with internet access and a contemporary browser. Thus, it is compatible with any desktop, tablet, or smartphone and is available to everyone.

Input Parameters:

Interactions with the bot shape how the system responds. All data flows through centralized servers for processing at scale.

System Outcomes: Responses pop up, links unlock, and alerts ping automatically - every move tied to what the system sees.

Constraints:

Without a stable web link or a phone that understands the app, things fall apart quickly for better or worse.

System Software Requirements Phase

Component	Specification / Description
Operating System	Windows, Linux, or macOS
Frontend Technologies	HTML, CSS, JavaScript
Backend Framework	Flask (Python)
Deployment Platform	Vercel (Frontend), Render (Backend)
Database	PostgreSQL

Table 5.2: System Software Requirements Phase

System Hardware Design Phase

1. Identify the functional blocks (web server, database server, user devices).
2. Create a process flow (user → games → backend → database → leaderboard) [1].
3. Find discrepancies (connectivity, response time).
4. Create user frontend, server backend, and database interfaces.
5. Perform system analysis for load and response time.
6. Create an integrated test plan for performance and network stability

System Software Design Phase

1. Clarify the different functional modules (Games, Authentication, Leader Board).
2. Illustrate workflow with UML diagrams and sequence charts.
3. Detect logical inconsistencies of module dependencies.
4. Create exterior interface points (REST APIs for data exchange).
5. Execute design-level analysis of algorithms and data handling.
6. Compose a detailed plan of unit and integration tests for all modules

5.2 Block Diagram

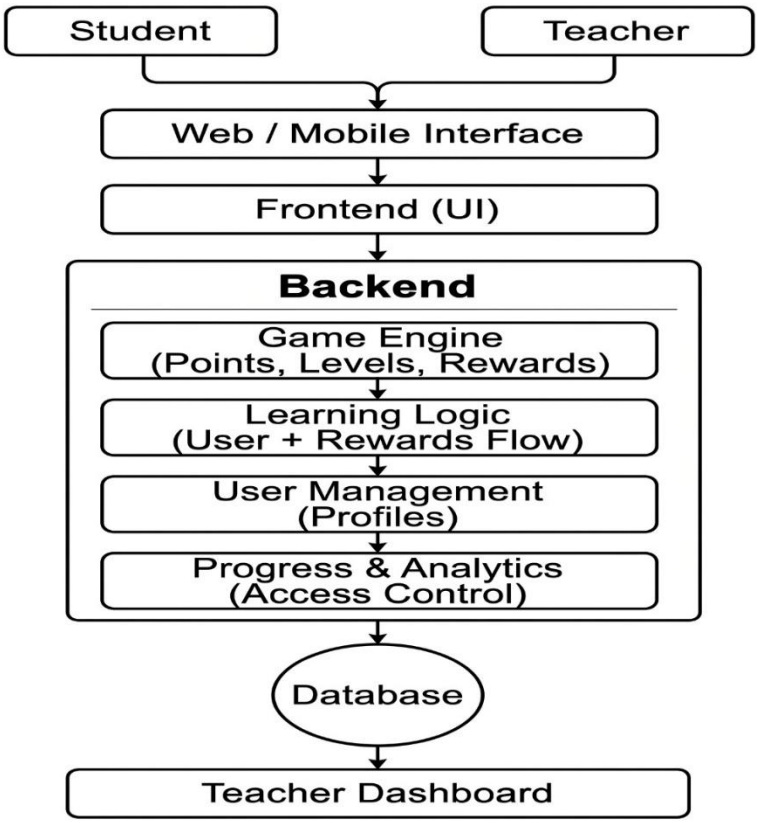

Fig 5.1 Functional block diagram

The front-end will consist of a user interface for users of the application. It has been built using HTML, CSS, and JavaScript to create an interactive dashboard for the users. The dashboard allows the user to:

- Register and log in
- Take quizzes
- View Scores and Leaderboards
- Select Languages

The front end of the application will also be built to be responsive, user-friendly, and able to be viewed on a variety of devices.

In this layer, all processing and business logic for the entire back-end of the application is carried out by the back-end of the application. In addition, it is also responsible for:

- Authentication of Users and Sessions

- Processing Requests for Quizzes
- Managing Communication Between Front End and Databases
- Providing APIs for Real-Time Updates

The Backend layer is responsible for handling data in a secure and efficient manner within the entire application.

Database All requisite data is secured in the database that allows the system to operate successfully:

- User Information
- Quiz Questions and Answers
- User Scores/Performance

The database will be relational so that the data remains consistent, retrieval is rapid, etc. Additionally, use of proper indexing and structured tables will enhance overall system performance.

The game engine is the main part which provides gamification to the platform. This module creates quiz questions, calculates scores and experience points (XP), determines difficulty (i.e., adjusts based on user performance), and gives users feedback in real time.

This module will allow learners to have an engaging and interactive experience while learning.With this functionality, users may utilize the system in any language by utilizing JSON format language files to alter the text displayed on the interface. This makes it possible for all users of differing backgrounds to experience the platform [1].

5.3 System Flow Chart

A graphical representation known as a system flow chart shows many operations and how data flows through a system. A system flow chart allows you to see how different pieces of the system connect to one another and how user inputs are processed to provide the resulting outputs or responses to the users.

For example, the system flow chart for the ShikshaSetu platform provides a visual representation of how all users interact with the ShikshaSetu system and how users use the

different modules within the ShikshaSetu system (for example, DB, Interactive Learning System, backend services, etc.) to provide those responses to users.

The system flow chart represents the entire execution flow of the ShikshaSetu system from the initial user interaction to the final generated response or output for the user.

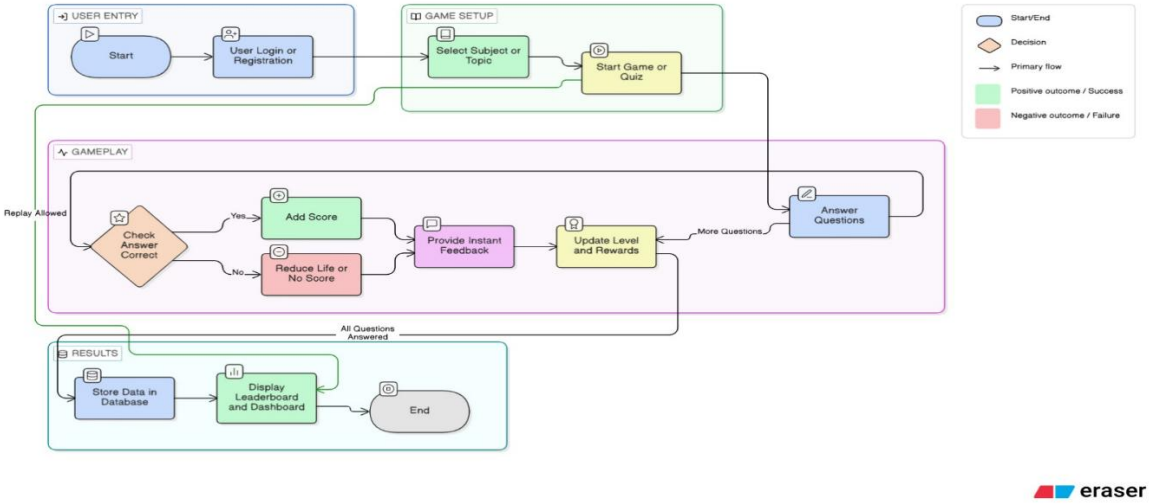

Fig 5.2 System flow chart

When someone enters the ShikshaSetu (ShikshaSetu is a Hindi word meaning "education bridge") to use it on their Internet-connected device (PC, tablet, or smartphone), they will see a Homepage (Home Page) containing menu options for Login, Registration, or Chatting with the Bot.

If the user does not have an Account with ShikshaSetu and wants to create one, he/she has to Register by providing their Name, Email Address, and Password; then the user has the ability to Login if the user has already registered.

If the user enters correct Login Information (credentials) into ShikshaSetu then they will be given access to ShikshaSetu; if not then the user will be prompted to try again. This workflow step assures ShikshaSetu that only individuals that have registered for an Account with the application will be given access to any of the features of the application.

The next workflow step is for an authenticated user with an Account to interactively use the ShikshaSetu application in any of the following ways:

- To access Academic Resources,
- To view Announcements and Updates,
- To Chat with the Bot, and
- To Track their Academic Performance.

In this way, Chatting with the Bot will be the primary mode of interacting with ShikshaSetu since users will use written, normal conversational Query/Request language/questions to Chat with the Bot and receive responses back from the Bot using written, normal conversational language double-amping the user's experience with ShikshaSetu.

The Interactive Learning System processes a user's input question based on its logic. This includes understanding what a user wants and determining how to respond. If the user's question matches a pattern, a response is provided; otherwise, the Interactive Learning System may give an alternative response or suggest possible responses. This phase is essential to ensure efficient and user-friendly interaction with the Interactive Learning System.

The Interactive Learning System app sends the user-requested information to the backend using a web-communication method. The back end (developed using Flask), has a set of functions to provide all the logic for the website and determine how to respond to the user's request. The back end, in addition to executing user requests, allows communication between the user and database and retrieves/stores data from/to the database. The back end of the application provides the required logic and data processing for the application.

Once the user has submitted their request, the Interactive Learning System app will retrieve data from or store data in the PostgreSQL database. Whenever a user wants to authenticate or personalize their account, the database will retrieve their personal records. The system will also request to retrieve academic resources based on the user's request. The database will also be updated with all user interaction with the Interactive Learning System app. These operations

allow for efficient and effective data management throughout the Interactive Learning System application.

All of the user's request has been processed and an appropriate response has been generated.

- Responses from the Interactive Learning System will be presented to the user.
- Any requested resource, information, or other data will be presented.
- The user will also see notifications or updates that may be relevant to their request.
- The response should be timely, clear and relevant to what the user requested.

Once the process is complete, the user will see their final output. The system should show the information in a way that is easy for the user to read and understand, and therefore easy to use.

After the user has completed an action, the system allows for an ongoing interaction where the user may perform multiple actions without having to start again from the beginning. Users may ask multiple questions and find different resources, and easily navigate the system.

The system process flow chart clearly shows how different types of users interact with different parts of the system at many levels within that same level. This functional breakdown of the database shows that all transactions can be processed in a very structured manner and will therefore allow users to receive accurate results and use the system efficiently.

5.4 Choosing Devices

Device selection is an important consideration for designing and implementing a system because the type of device chosen will affect how well a system performs, how reliable it is and what level of accessibility will be provided to the user through the devices chosen. The ShikshaSetu system was designed with the objective of ensuring seamless operation, cost-effectiveness and compatibility with current web technologies as device selection.

Because the system proposed is a web-based academic platform, it does not require special hardware components or embedded systems to operate; therefore it is designed to operate efficiently on general computing devices typically used by users. This means that the ShikshaSetu system can be used by as many people as possible, without requiring users to purchase additional hardware.

Device selection considerations have focused on supporting both front-end user interaction and back-end operations of the system so that all parts of the overall system can interact and perform correctly.

Component	Options Evaluated	Final Choice	Reason
Backend Framework	Django, Node.js, Flask	Flask	Lightweight, flexible, suitable for Interactive Learning System workflows.
Database	MySQL, MongoDB, PostgreSQL	PostgreSQL	Reliable for Leader Board.
Frontend	React, Vue, HTML/CSS/JS	HTML/CSS/JS	Simplest for deployment and responsiveness.
Deployment	Netlify, Heroku, Vercel/Render	Both	Ensures global scalability.

Table 5.3 Software Components

Backend

The backend of an application handles all system functionality, user requests, and data management operations. The ShikshaSetu system uses several different back-end infrastructures:

- **Cloud servers:** The cloud services (for example, Render) hosting the ShikshaSetu system provide scalable, reliable server environments for supporting the application logic, processing user requests, and facilitating communication between the various components of the system.
- **Database servers:** The PostgreSQL database will provide a complete database management solution that supports the data for all of the modules of the system, including user data (e.g., profile information), resources (e.g., academic academic programs), and user interactions (e.g., Interactive Learning Systems). The database server provides efficient storage, retrieval, and integrity for the system's data.

Using cloud-based back-end systems removes the need to maintain physical servers and also provides the ShikshaSetu system with high availability and scalability.

Frontend:

Although modern JavaScript frameworks such as React and Vue provide excellent UI capabilities, the project decided to use standard HTML, CSS, and JavaScript. By making this choice, the development process is simplified, complicated build configurations are avoided, and it is ensured that the interface loads quickly on different devices. Furthermore, the selected stack is simpler in terms of deployment and maintenance.

Deployment:

Netlify, Heroku, and Vercel/Render have been compared in terms of reliability, performance, and cost. The last configuration is using Vercel as well as Render—Vercel is handling the frontend, thus a user can get a fast global delivery, and Render is hosting the backend and PostgreSQL database. So, this duo is giving a safe and scalable space with the capacity to handle live conversations with the Interactive Learning System, booking activities, and encrypted transactions

5.5 Designing Units

The design of the single unit of a system is one of the most important steps within the overall systems development process. The overall systems development process consists of creating the external and internal designs of each unit/part/component that make up the entire system. The ShikshaSetu platform consists of several functional units, with each unit having a defined responsibility.

Each of the functional units within the ShikshaSetu platform has been designed in such a way as to provide maximum modularity, scalability, and ease of future maintenance. The system follows a modular architecture by having a clearly defined role for each individual unit and by having each unit communicate with other units through defined interfaces. This allows for easier development, testing, and enhancements in the future.

5.6 Standards

The development of Shiksha Setu system relies on standards, which are really important to the quality and development of any software system. They ensure consistency between software

systems, ensure interoperability, ensure security, and ensure quality in the software.When Shiksha Setu was developed, a number of industry and web standards were used to develop the platform. This has allowed Shiksha Setu to be reliable, scalable, and able to interact with various technologies and devices through the use of a number of standards.

These standards give Shiksha Setu the framework to develop system architecture, communication protocols, data handling, and security mechanisms. By following standards, Shiksha Setu can provide a better performance, better maintainability, and better user experience for end users.

Standard Type	**Examples**	**Relevance**
Communication	HTTPS, REST	Secure data flow
Data Format	JSON	Multilingual responses
Security	Password hashing, TLS	Secure user authentication
Architecture	ISO/IEC 30141	Cloud-based deployment structure

Table 5.4 Standards

Web Standards

The ShikshaSetu application has been developed for use on the World Wide Web and uses web technologies to remain compatible and responsive [4].

- HTML - The Hyper Text Markup Language (HTML) is the markup language used to structure the content of web pages. HTML provides the framework for how the User Interface (UI) is laid out and what elements are used to build the UI.
- CSS - Cascading Style Sheets (CSS) are used to style and design the user interface (UI). CSS is used to standardise how the layout, colours, and responsiveness are applied to the UI across devices.
- JavaScript - JavaScript is used to provide interactivity and dynamic functionality to the system. JavaScript is used to provide instant updates to the user and support the user in interacting with the system.

These web-based standards ensure that the system will work reliably with modern web browsers and provide a consistent user experience.

Communication Standards

This system has been designed and built to use standard protocols for communication between the various components (frontend, backend and the database) in order to facilitate communication.

- HTTP/HTTPS Protocol - The Hypertext Transfer Protocol (HTTP) is the protocol used to communicate between a client and a server. HTTPS is an extension of HTTP that provides security through a secure connection via encryption of the data being transmitted [10].
- RESTful API Architecture - The RESTful API Architecture is the design of the system to use REST principles for data communication. Data is exchanged as part of the standard set of HTTP methods, which include GET, POST, PUT and DELETE.
- JSON - JavaScript Object Notation (JSON) is the data exchange format that is used for the exchange of data between the front and back ends of the system because of its lightweight and ease of parsing.

These communication standards highlight how commonly accepted standards for the transfer of data are used within the application to ensure that the transfer of data takes place quickly, easily and securely [7].

Data Format Standards

Relational Databases are used as a reference for the database design.

- By using a relational database, the data is organized through tables with predefined and ruled relationships so the data can be kept consistent and dependable.
- Data is entered into the PostgreSQL database through SQL (Structured Query Language) to extract and manage the data.
- Normalization is another way the database design is structured by reducing the duplicate data, thus making the database system more efficient.

These standards will ensure the database can support a continued growth, be able to rely on its operations, and be easy to support.

Security Standards

Security is equally important to the overall operation, especially when it comes to handling user authentication and information.

- Authentication/Authorized use of the system and granting access permission is restricted by implementing a system of ROLE BASED ACCESS CONTROL.
- All user passwords are stored securely through an encryption scheme that prevents others from accessing the user's password.
- All communications between the client and the server (Web) is secured using HTTPS ENCRYPTION.

User sessions are securely managed to prohibit unauthorized access to the system. These security standards will provide reliable security to user information and ensure system reliability.

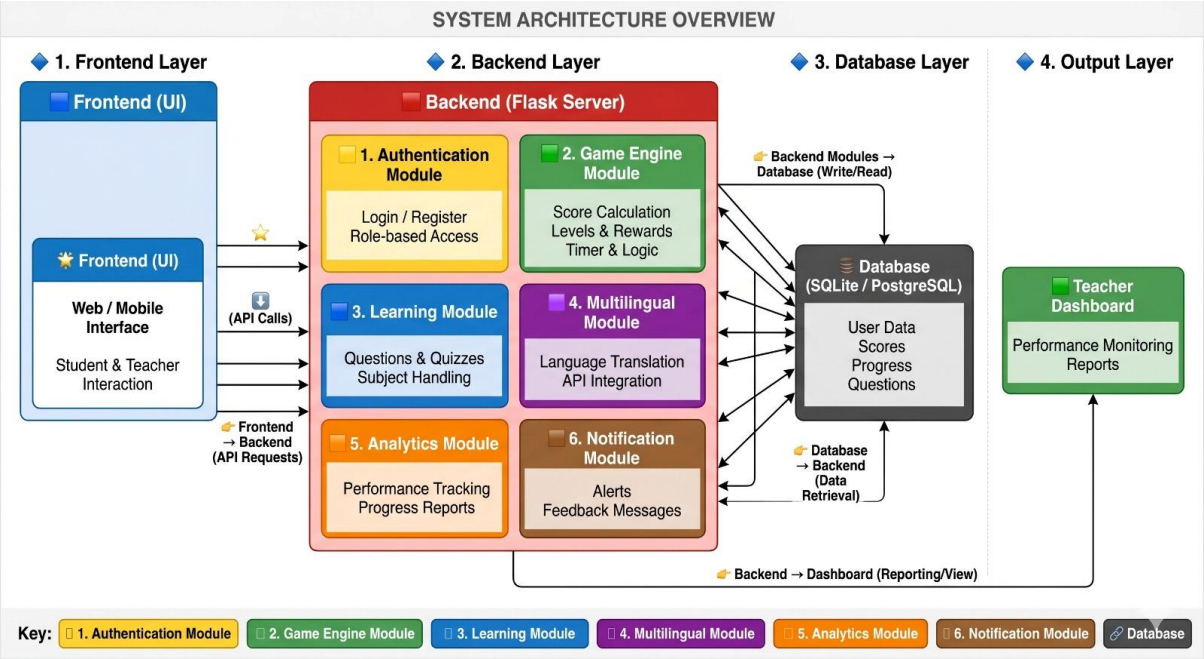

Fig 5.3 Backend system design

Architectural and Interoperability Standards

The architecture of ShikshaSetu system is based on a multi-layered design that brings together user interaction, processing, and data management. The architecture facilitates seamless communication among various system components and at the same time, keeps the system scalable and efficient.

Basically, the whole system is composed of three primary layers: Frontend (User Interface), Backend Processing Layer, and Database Layer. These parts interact with each other to offer an engaging and game-based learning experience.

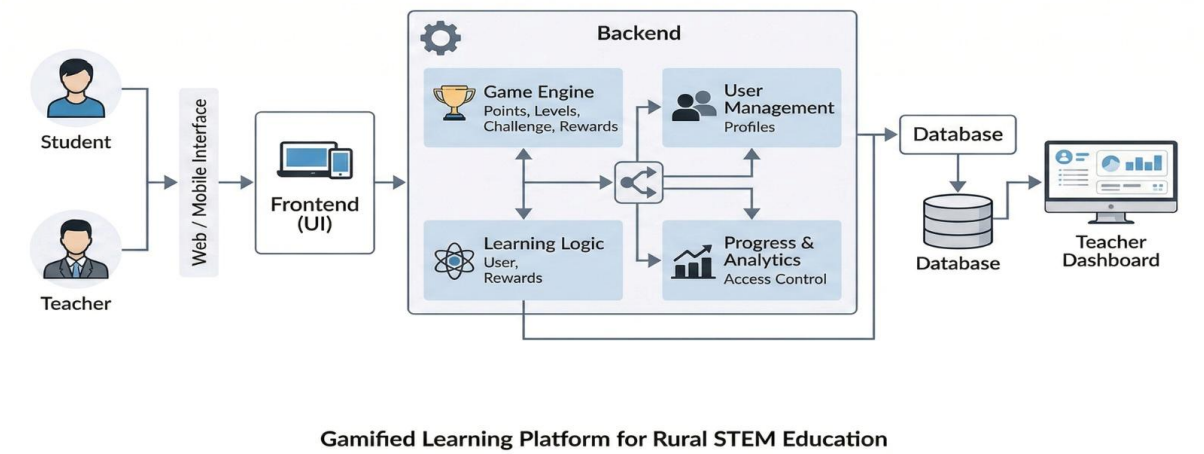

Fig 5.4 System Architecture

5.7 Mapping with IoTWF Reference Model Layers

The Internet of Things World Forum Reference Model (IoT WFRM) is a global standard for organizations creating both applications and systems intended to process information via the internet and/or provide real-time services to any technology that connects (sensors, devices, etc.) to the internet using various layer models (software, hardware, network, etc.).

Although ShikshaSetu is an academic web site (not an IoT-enabled device) with its own separate design, conceptually, the ShikshaSetu architecture maps to the IoT WFRM model (except for layer 3 - data continuity, as there are no data continuity requirements through ShikshaSetu).

This mapping provides a better understanding of how information flows through the layers of ShikshaSetu, how it is processed, and how users interface with it. In addition, by aligning to the IoT WFRM model, ShikshaSetu design methodology is now more formalized in terms of using a layered analysis vs. other non-layered approaches.

Layer	Project Component
7. Processes	Academic workflows such as resource sharing and communication
6. Application	Web-based user interface
5. Data Abstraction	Flask routing, JSON
4. Data Accumulation	PostgreSQL
3. Edge Computing	Backend logic processing
2. Connectivity	HTTPS requests
1. Devices	User smartphones/PC

Table 5.5 Mapping Project layers with IoT WFRM

Layer 1: Devices

The physical devices are represented in the Physical Devices Layer of the ShikshaSetu system's Interaction with the System. User devices used by both students, faculty and administrators in order to connect to and interact with the platform are represented in this layer including all desktop/laptop/smartphone/tablet user devices.

Layer 2: Connectivity

The layer of connectivity enables communication among devices and the system. In this project, connectivity is achieved by various means such as:

- The use of Internet Protocols (Wi-Fi and mobile data)
- The use of HTTP or HTTPS to communicate between the client and server
- This layer between the user and back-end will transmit a user request to the back end and return a response to the user.

Layer 3: Edge Computing

The edge computing layer is where initial data processing is completed closest to where it was generated. In the ShikshaSetu system, there are two elements to this layer:

- Client-side processing using Javascript
- Input validation and basic data processing on the front end.

By performing simple operations at the user device's level, this layer reduces the burden on the server.

Layer 4: Data Accumulation

The data storage layer is responsible for holding all of the data that was generated by the system. In the ShikshaSetu system, data accumulation will take place in the following manner:

- By using a PostgreSQL database
- By storing user data, academic resources and system logs

This layer will allow for the efficient collection and storage for additional processing of data.

Layer 5: Data Abstraction

The data abstraction layer processes and organizes all types of stored data so that it can be used meaningfully. In the ShikshaSetu system, this includes back end processing via Flask, API handling and formatting, and raw data conversion into structured responses.Transformation of the data into a usable format for the Application Layer occurs within this layer.

Layer 6: Application

The Application Layer is where users interact with the system and is the end-user interface. For this project, this layer includes a web-based user interface with dashboards and resource pages as well as user modules. Functionality for accessing academic resources, receiving updates from the system, and interacting with the system are provided by this layer.

Layer 7: Processes

The topmost layer is the System Integration Layer, which connects the system to real-world processes and user activities. For the ShikshaSetu System, this layer includes academic workflows (resource sharing and communication) and user interactions with one another, as well as decision-making processes based on data in the system. This layer allows the system to efficiently provide support for real-world educational activities.

5.8 Domain Model Specification

A Domain Model illustrates how one's Organization looks in terms of conceptualizing the data in the Organization or system by showing the primary entities and attributes plus the relationship among these entities. This is the overarching and high-level representation of how the data is logically organized within the Organization/System and how these logical parts interact with each other.

For example, in the case of the ShikshaSetu system, the Domain Model contains the three main entities in the academic management domain and show how they are organized and related to one another: 1) Users, 2) Resources, and 3) System/User operations. The Domain Model provides a logical overview of the system and will serve as a basis for database design and system implementation.

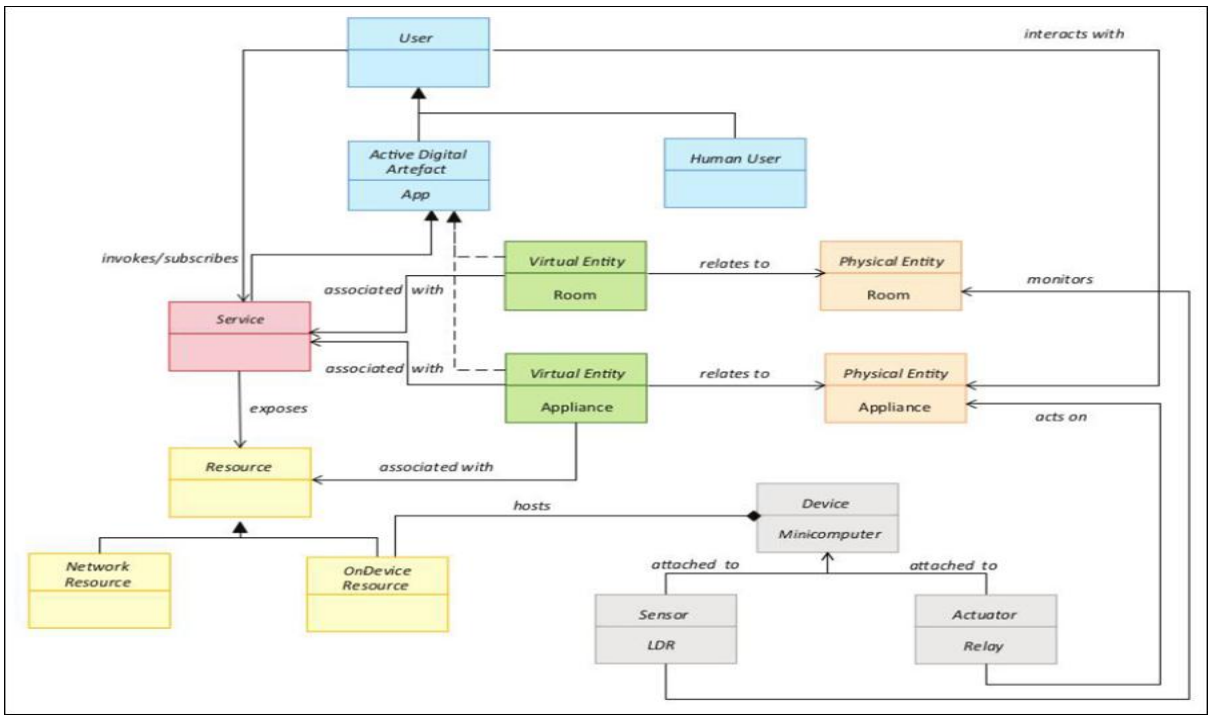

Fig 5.5 Domain model for Home Automation

1. User Entity

User entity covers all people using the system, i.e. students, faculty, and administrators. User entity has the following attributes:

- User ID
- User Name
- User Email
- User Password
- User Role (Student / Faculty / Admin)

User entity's functions are as follows:

- Provides access to the system
- Allows users to utilize the features of the system
- Provides functionality to allow a user to perform the responsibilities of their assigned role

2. Admin Entity

Admin entity is responsible for system management and operation control. Admin entity has the following attributes:

- Admin ID
- Admin Name
- Admin Email
- Admin Role

Admin entity's functions are as follows:

- Manage Users and Resource
- Update Content in the System
- Monitor Activity in the System

3. Academic Resource Entity

The Academic Resource entity describes all materials (education) available within the system. Academic Resource entity has the following attributes:

- Resource ID
- Resource Title
- Resource Description
- Document/Resource

- Uploaded By
- Upload Date

Academic Resource entity's functions are as follows:

- Store all study materials and academic resources
- Provide access to all users through the system
- Support the function of content management for the system

4. Notification Entity

The Notification entity describes all messages and updates generated from the system. Notification entity has the following attributes:

- Notification ID
- Notification Message
- Notification Date

Notification Targeted User Notification entity's functions are as follows:

- Send notifications to targeted users
- Enhance internal communication between users of the system

5. Authentication Entity

Authentication Entity is responsible for logging into the system and security data. Authentication Entity has the following attributes:

- Login ID
- Login Name (User Name)
- Password

Role Authentication entity's functions are as follows:

- Authentication of user credentials
- Authentication control of users to features of the system

5.9 Communication Model

A communication model is a representation of how any variable in the system can be communicated to the other components in the system in a seamless manner.

The communication model defining the ShikshaSetu platform consists of two parts: the communication of the user interface or client component to the backend or server component, and the communication of the backend/server component to the database or data storage component.

In this model, the communication architecture uses a client-server or two-way communication method where the client communicates with the server, and the server communicates with the database or data storage. This structured communication architecture provides for ease of use, expansion.

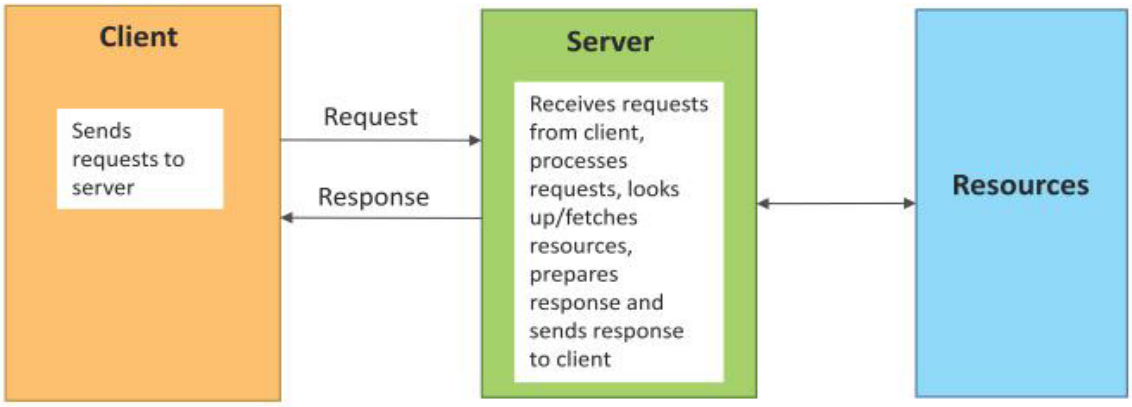

Fig 5.6 Communication model suitable for Home Automation

Types of communication

1. Communication of Request

In this case, the user sends information to the system.

- E.g.: logon, access resources, submit data
- Flow (direction): client → frontend → backend

2. Communication of Response

In this case, the system sends information back to the user.

- E.g.: display resources, show results
- Flow (direction): backend → frontend → client

3. Communication of Database

In this case, the backend communicates with the database.

- E.g.: get user's data, save records
- Flow: backend ↔ database

Characteristics of the Communication Model for ShikshaSetu

- Reliability: Data transmission is always accurate
- Security: HTTPS is used for all communications via encrypted websites.
- Scalability: System can handle many users at the same time.
- Efficiency: Request/Response cycle is as short a time period as possible.

5.10 IoT Deployment Level

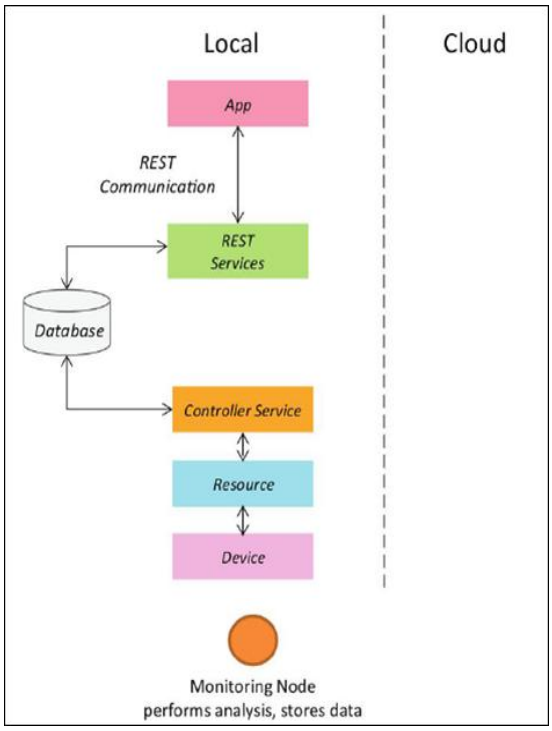

Fig 5.7 IoT deployment level suitable for Home Automation

Defining the IoT deployment level will help you understand how your system is organized regarding creating, processing, and communicating data between components. Each level describes a different type of system, considering factors such as complexity, scale, and communication architecture. Although the ShikshaSetu system is mainly a web-based academic platform, mapping it to an appropriate IoT deployment level will allow for a better understanding of its architecture and data movement.

The ShikshaSetu system operates using a cloud-based client-server model closely corresponding to IoT level 3. Therefore, using multiple devices to communicate to the cloud for processing and managing your data will help ensure that your ShikshaSetu system is set up correctly.

5.11 Functional View

The system can be detailed and understood through the functional view in relation to its functionality, how functions/modules are connected to one another. It will demonstrate the function(s) of the system and describe how the system's different components will work together to achieve the overall objectives of the system based on their functionality.

On the ShikshaSetu platform, the functional view provides a complete visual representation of the processes that will provide the user with a motionless experience through the flow of operations among the individual modules (i.e., user management, resource management, authentication, and messaging) and guarantee that all aspects of fulfilling the system requirements are completed.

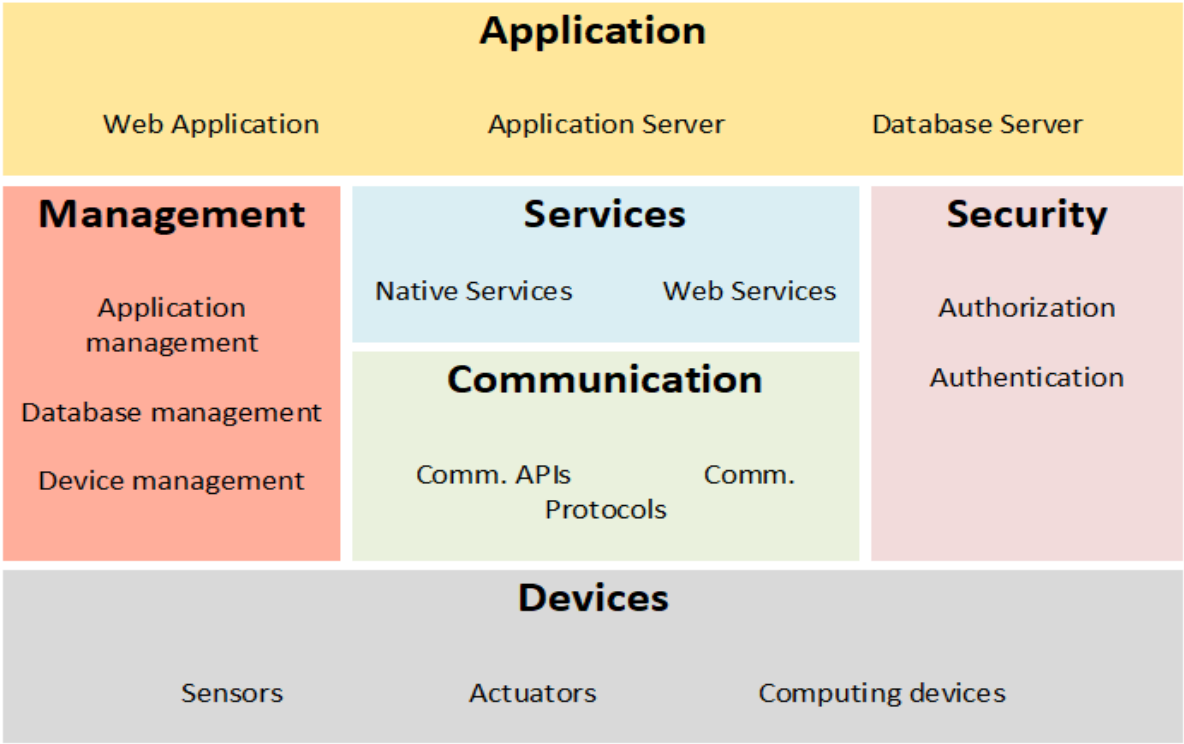

Fig 5.8 Functional view for Home Automation

1. User Administration Functionality

The function of user administration is to control all operations that consist of user interactions with the system.

Types of interaction/Tasks

- User Registrations and Login
- User Profile Management
- User Access to the system based on the user's role.

Functionality

- Allowing users to establish their own accounts
- Maintaining all user data
- Providing users role based Access

2. User Authentication Functionality

The function of user authentication provides a means of ensuring that the system is only accessed by those customers that have been approved to use the system.

Types of Tasks

- Verification of customers' login credentials
- Management of a customer's created session while using the system
- Access control to the system

Functionality

- Validates a user’s identity
- Provides protection to a user’s information found in the system.
- Prevents unauthorised individuals from accessing the system.

3. Resource Management Functionality

The function of resource management includes the management of academic resources that are stored within the system.

Types of Tasks

- Uploading of Study Material
- Updating/Deleting Resources
- Accessing Academic Content

Functionality

- Providing customers (the Faculty) with a central location to access their resources
- Allowing Faculty to manage academic resources that are uploaded to the system
- Provides students with a means to access/obtain academic resources.

4. Data Processing Functionality

The function of Data Processing within the system completes the main (Core) processing that is completed on a customer's request**.**

Types of Tasks

- Processing User Input
- Back end Logic Execution
- Response Generation

Functionality

- Connect Users (Front-End) to Database (Back-End)
- Provides efficient processing of Data
- Ensures the correct user output will be produced.

5. Notification Function

Notifications are the means by which the users of the app and system are able to communicate with one another.

Types of Tasks

- Sending updates and/or announcements
- Providing users with notifications of new activity.

Functionality

- Informing users of what is happening in the system.
- Allowing for improved communication within the platform.

6. Database Management Function

The database management function will manage the recording, storing, and accessing of the information for the users of the app and system [10].

Types of Tasks

- User information
- User resources
- System logs

Functionality

- Ensure the integrity and consistency of information in the database.
- Support the efficient operation of the database.

5.12 Mapping Deployment Level With Functional Blocks

The operational view specification describes the operational aspects or "how things run" of a system's execution in a real-life environment. This pertains to the dynamic nature of a system, including how different components will interact at runtime (the runtime interaction of different components), the flow of data from module to module, and how user requests are processed to produce outputs.

In the case of ShikshaSetu, the operational view outlines how the ShikshaSetu platform operates from the perspective of users using it. This provides clarity about how the system executes; i.e., what users do, processing the user actions/requests through the back-end, how back-end processing works with the database, and generating responses to the user. Understanding the operational view of a system is critical for evaluating system performance, functional reliability, as well as evaluating system behavior in real-time.

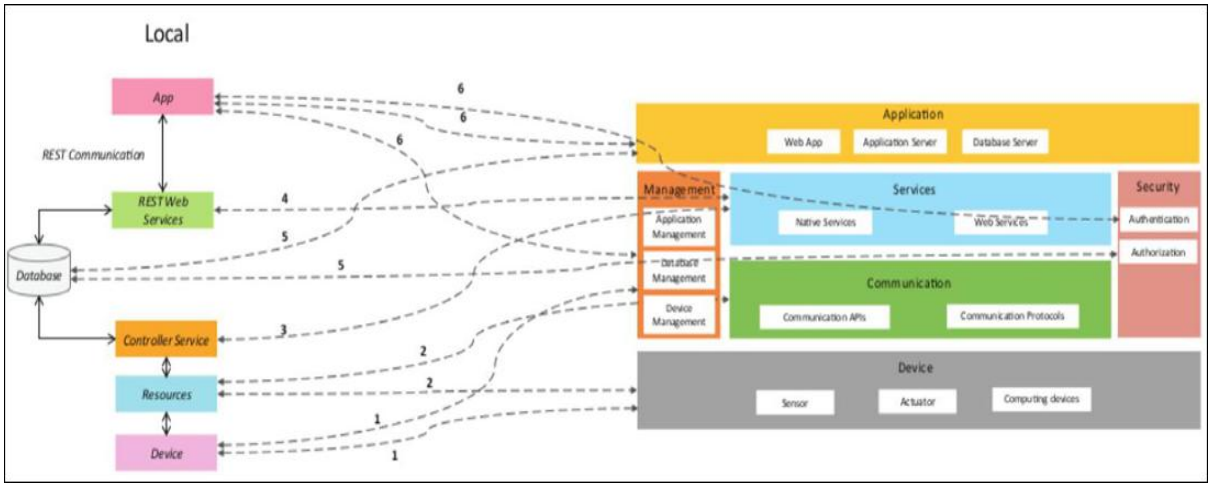

Fig 5.9 mapping IoT deployment level with functional view

Device Layer → (User Interface Level)

This is the layer that consists of all devices that can be used by the end user including tablets, smartphones, laptops (desktop computers) and anything else similar in terms of design or function, directly connecting to the front end of the system.

Functional Mapping:

- User interaction (logging in, taking quizzes),
- Displaying dashboards, scores, and leaderboard, etc.
- Choosing the language and navigating through the system.

The purpose of this layer is for users to be able to have easy access and to use the system via a responsive interface.

Communication Layer

This tier serves the purpose of transmitting data from the user interface to the backend system.

Functional Mapping:

- Send requests from users (e.g. login, quiz attempts)
- Receive server responses
- Communication using APIs between frontend and backend

Effective communication provides real-time updates to users and enhances the overall performance of the system.

Processing Layer (Application Layer)

This tier consists of the backend server and the game engine which perform all logical operations.

Functional Mapping:

- Authenticate and maintain user sessions
- Generate and evaluate quizzes
- Calculate scores and assign XP
- Adjust difficulty based on user performance

This is where most of the intelligence and decision-making processes within the system occur.

Data Layer (Database Layer)

This tier is responsible for managing the storage and retrieval of data.

Functional Mapping:

- Store user data
- Manage quizzes and associated results
- Maintain performance history

The database maintains data consistency, reliability, and facilitates fast retrieval of information.

Analytics and Visualization Layer

This tier produces insight into user performance.

Functional Mapping:

- Generate leaderboards
- Track and report on performance
- Visualize feedback

It provides a way for both users and admins to understand their progression as well as the effectiveness of the system [8].

5.13 Operational view

The information view specification details how information is structured, organized and stored in a system. The key components of the information view are relationships between various data elements and how these relationships define the data flow through the system.

In the ShikshaSetu system, the information view defines how academic data, user information and data collected about the system are organized to enable efficient processing of this data. It also defines how the structure of data supports system functionality, scalability and integrity.

Using a systematic method for managing data, the system employs a relational database to store information. This database contains many tables that contain information about different entities, such as users, resources, and notifications.

Data is placed in the database according to a strict design so that when the data is retrieved or modified, it will be done so efficiently, orderly and reliably.

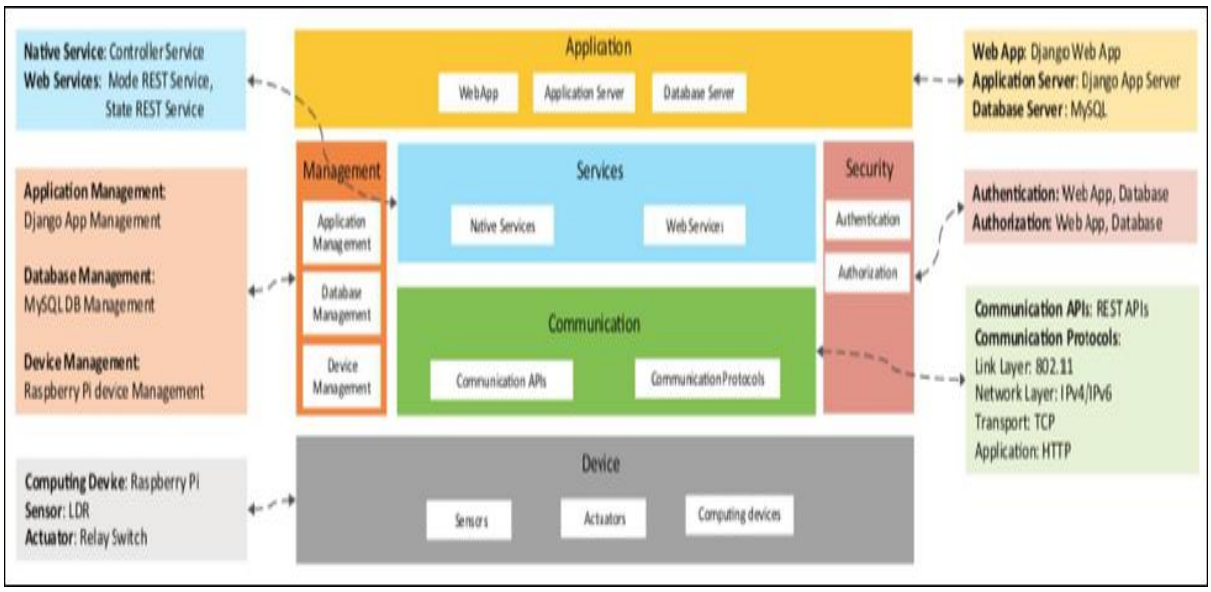

Fig 5.10 Operational view

The operational view of a System outlines how a system will operate when used in live form as well as how the various components will work together to accomplish the objective of the system. This section will be focused primarily on how users will be able to operate the system and provide details on what processes will occur when a user makes a request from the system. The business user will begin by logging into the System. This action is performed through the front-end interface in conjunction with the back-end server through an API call. Once the back-end server has received this request, it processes the user input and performs the necessary actions. For example, when a user begins a quiz, the database will query the database for the quiz questions and return them to the user via the front-end interface. When the user submits

his/her answers to the quiz questions, the data from the quiz submission will be sent from the user to the back-end server to be evaluated via the game engine to determine the score and experience points (XP) earned for the quiz.

The database is critical to the operation of the System as it is responsible for storing and retrieving data throughout operation of the System. The database will store user information, quizzes and user performance. The database will keep track of all of the actions performed by a user, so that the user can be assured of the integrity and accuracy of the data that he/she has in the database.

In addition to being event-driven architecture that is specifically designed for every user action to trigger some sort of result on the application, this allows the application to be able to provide real-time updates such as score changes, leaderboard updates and feedback messages immediately after they happen.

Furthermore, the system can support multiple users that are accessing the system at the same time. Proper session management allows each user's information to be stored and maintained independently so that no other users are impacted. The system also employs security measures, such as authentication and validation to protect the integrity of each user's information and to prevent unauthorized access to user data.

The workflow of the system is summarized below:

- The user accesses the system via the front-end interface.
- The user performs an action, which is submitted to the back-end as a request.
- The back-end receives the request, processes the request and interacts with the database.
- The game engine evaluates all responses received from the request and updates all scores.
- The results are sent back from the back-end to the front-end and displayed to the user.

This structure of operation for the system allows for the system to be operated in a smooth manner, for all the data handled within the system operate efficiently and for each user to have a responsive experience while using the system. Also, this structure of operation allows the system to scale, as the number of users increases, without experiencing performance issues due to numerous changes being made to the system.

5.14 Other Design Aspects

The deployment view specification defines in detail how a system will be physically set up and distributed across multiple physical hardware and software environments. This includes the system component configuration, where they'll be located, and how they interact with each other under real-world conditions.

The ShikshaSetu system implements deployment through a cloud-based architecture that provides scalability, reliability, and availability.

1. User Interface Development

The UI was designed intuitively, simply and with ease of navigation, in mind; layout utilizing dashboard, quiz panel and leaderboard visual elements are organized in a clear manner; colors, icons and animations are used in ways that increase user participation, and create a more interactive way for users to learn.

2. Modular Design System

The system follows a modular design approach. Therefore all components of the system operate independently from one another, e.g.; Frontend (UIS), Backend (API), Database and Game Engine. This method allows for easy maintenance, error checking/repairing and enhancment in the future without affecting the other parts of your system.

3. Security Setup

Security is important to the platform. User accounts were implemented with authentication mechanisms in place to only allow authenticated users access to the platform. Sensitive data such as user info and score data are securely stored and have validation specified in order to prevent acccess to unauthorized users.

4. Performance Optimization

The platform was created to respond quickly and run smoothly. Database Querying is efficient/fast and Backend Processing is optimally performed to minimize delay time. Also, frontend to backend async communication allows for real-time updates of page contents, without needing to refresh the page.

5. Scalability

The architecture of the platform accommodates growth of users based on the use of lightweight frameworks and the design structure of the database can accommodate the system without major structural modifications.

6. Multi-language

The platform has a multi-language capability to allow the system to support users from various areas. The platform supports an interface that changes based on dynamic language files which allow users to access the platform with a larger amount of users.

7. Error handling and Validation

The system has developed an implemented an appropriate mechanism for handling errors and validating inputs, therefore, only valid data will be processed, which will help to reduce data integrity issues and provide a more reliable system.

8. Maintenance

The coding within the platform is modularly organized and thus creates the opportunity for a developer to update, modify, and improve the system in the future.

Conclusion

Overall, the design-related factors discussed above improve on the quality of the system's usability, security, and performance. Therefore, ensuring that the platform is reliable, scalable, and will operate successfully with the real-world's user base.

CHAPTER 6

HARDWARE, SOFTWARE AND SIMULATION

6.1 Hardware

Since the proposed system is a web application, it does not mainly depend on specific hardware components. For development, deployment, and user interaction, a basic computing device will be sufficient.

Hardware	Purpose	Description
Developer Laptop/PC	Coding, testing, simulation	Used for installing required software, executing Flask backend, running local servers, and performing debugging. Standard specifications include ≥8GB RAM, multi-core CPU, and stable internet connectivity.
Smartphones	UI testing	Used to test mobile-responsiveness of the multilingual interface and Games.

Table 6.1 Hardware Components Used

Though no physical IoT devices are involved, the hardware components are interacting in a layered cloud architecture:

1. Frontend users (mobile/desktop) → send requests to
2. Vercel Edge Servers → which route requests to
3. Flask Backend on Render → which, in turn, gets or saves the data in
4. PostgreSQL Cloud Database.

Such a cloud-centric architecture is giving way to an always-on service, language-diverse access, and traffic of a Gaming Scale, which have been elaborated in the sections about the architecture and the implementation of the project.

6.2 Software Development Tools

The whole system was created with contemporary, open-source software tools that are capable of supporting web applications, Interactive Learning Systems, databases, and deployment pipelines.

Programming Languages

Language	Usage
Python	Used for backend development and server-side logic.
JavaScript	Used for dynamic frontend interactions
HTML5/CSS3	Core UI design, responsive layout for cross-device support.

Table 6.2 Programming Languages

Backend Development Tools

Tool	Purpose
Flask (Python)	A lightweight Python framework for developing APIs, processing requests, and handling server-side activities. It allows you to split your code into modules and is pretty straightforward to link with databases.
Database	Used for storing user details, quiz data, scores, and progress. It ensures efficient data retrieval and consistency.

Table 6.3 Backend Development Tools

Frontend Development Tools

Tool	Purpose
HTML5	Structure of web pages
CSS3	Styling and layout design.
JavaScript Fetch API	Interactive elements such as quizzes, animations, and real-time updates.

Table 6.4 Frontend Development Tools

6.3 Software Code

User Login (Flask Backend)

```
from flask import Flask, request, jsonify
app = Flask(__name__)
users = {"student1": "1234"}
```

```
@app.route('/login', methods=['POST'])
def login():
    data = request.json
    username = data.get("username")
    password = data.get("password")

    if username in users and users[username] == password:
        return jsonify({"message": "Login Successful"})
    else:
        return jsonify({"message": "Invalid Credentials"})

if __name__ == '__main__':
    app.run(debug=True)
```

Explanation:

The code above manages user authentication in Flask framework. It collects user's login information via POST request and checks that the username and password are correct. When the input username and password are the same as those stored, the server responds with a message indicating success, else it sends an error message. This way, only legitimate users can get into the site, leading to better security.

Quiz Question Fetching

```
@app.route('/get_question', methods=['GET'])
def get_question():
    question = {
        "question": "What is 2 + 2?",
        "options": ["2", "3", "4", "5"],
        "answer": "4"
    }
    return jsonify(question)
```

Explanation:

This code is the source of quiz questions for the user. The frontend sends a request and the backend returns a question and several options in JSON. Formats. In this way, the quiz content can be loaded dynamically without the page being refreshed. It's a starting point for interactive educational activities.

Score Calculation (Game Engine Logic)

```
def calculate_score(user_answer, correct_answer):
    if user_answer == correct_answer:
        return 10
    else:
        return 0
```

Explanation:

This function awarding score depending user answer. When user answer is same as the correct one, user get points (XP). If answer is not correct, then user get no points. This basic game mechanic is a part of the gamification system, which motivates users to improve themselves by rewarding accurate answers.

Leaderboard Logic

```
leaderboard = []
def update_leaderboard(username, score):
    leaderboard.append({"user": username, "score": score})
    leaderboard.sort(key=lambda x: x['score'], reverse=True)
    return leaderboard
```

Explaination:

This script keeps up a ranking board showing who scores the best. When a user finishes a quiz their winning number is put to the list, then the list is reversed. This stuff brings rivalry between users and pushes them to do better.

Frontend HTML (index.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>Quiz</title>
</head>
<body>
<h2 id="question"></h2>
<button onclick="checkAnswer('2')">2</button>
<button onclick="checkAnswer('3')">3</button>
<button onclick="checkAnswer('4')">4</button>
<button onclick="checkAnswer('5')">5</button>
<script>
  let correctAnswer = "4";
  document.getElementById("question").innerText = "What is 2 + 2?";
  function checkAnswer(ans) {
    if (ans === correctAnswer) {
      alert("Correct! +10 XP");
    } else {
      alert("Wrong Answer");
    }
  }
</script>
</body>
</html>
```

Explanation:

Here is the frontend code for the quiz interface. It shows a question and multiple answer choices. After the user picks a choice, the system verifies if the selected answer is the right one and gives them feedback immediately. This instant interaction boosts the user's involvement and turns learning into a more interactive experience.

Multilingual Support (JSON Example)

```
{
  "en": {
    "welcome": "Welcome",
    "start": "Start Quiz"
  },
  "kn": {
    "welcome": "ಸ್ವಾಗತ",
    "start": "ಕ್ವಿಜ್ ಪ್ರಾರಂಭಿಸಿ"
  }
}
```

Explanation:

Multilingual capabilities in the system are achieved through this JSON file. It keeps text in various languages and changes the UI accordingly when the user picks a different language. This feature is very handy for making the platform accessible to users of different language backgrounds without any difficulty.

6.4 Simulation

Simulation is a vital stage in the life cycle of a system development. It is when the software is run inside a set-up that mimics the real environment, in order to see if things work correctly before the system is actually made available to the users.

ShikshaSetu platform was simulated in this project through a local development environment. The main server was started with the help of a lightweight web application framework called Flask, while the frontend user interface was made available using a standard web browser. This arrangement provided a way for the components of the system to communicate and work together live.

Simulation Process

The simulation starts when user contact is made via the frontend interface. Users have an option to sign up, to sign in, and to choose quizzes. After a quiz has been chosen, the message is forwarded to the backend server that resposnds to the message and gets appropriate questions from the database [4].

When the user answers question after question, the system checks the answer against the game engine logic. Depending on the answer being right or wrong, the scores and experience points (XP) are determined. The results are even shown on the user interface, so the user gets instant feedback.

Test Scenarios

The following scenarios were tested during simulation:

- User registration and login functionality
- Quiz loading and question display
- Answer submission and score calculation
- Leaderboard updates based on performance
- Multilingual switching of the interface
- Error handling for invalid inputs

Performance Evaluation

In the course of simulation, the system was assessed considering the following aspects:

Response Time: The system reacts swiftly to the user's moves without any noticeable lag **Accuracy:** The quiz checking and the calculation of the final score are always correct and in line with the quiz Evaluation

Usability: The layout is easy to operate and user-friendly Reliability: There are no crashes, and the performance remains consistent

Simulation Results

The simulation results show that the system is capable of working as intended. Each component is operating right, and the integration of frontend, backend, and database has been successful. The platform can handle users' input efficiently and offers them immediate feedback [9].

Gamification components like awarding points and updating the leaderboard increase user engagement significantly. Besides that, the multilingual option functions properly, making it possible for users to change their language without compromising the system's performance.

CHAPTER 7

EVALUATION AND RESULTS

7.1 Test Points

ShikshaSetu system has been tested at several crucial points to ensure that all functions are running smoothly and efficiently. Each system module was not only tested separately but also in combination with other components during the integration tests. The main testing points discovered in the system were:

1. User Authentication Module

- Verification of login and registration functionalities to ensure system access security.

2. Frontend Interface

- Conducting user interface responsiveness and usability testing for a variety of devices such as desktops, laptops, and mobiles.

3. Backend Processing

- Making sure that user requests are processed correctly and that proper responses are given.

4. Database Operations

- Checking whether data is properly stored, retrieved, and updated in the PostgreSQL database [10].

5. API Communication

- Testing the frontend and backend communication mechanisms through use of API calls.

6. Deployment Functionality

- Making sure that the system is efficiently hosted and can be reached via cloud platforms.

All these testing aspects were scrutinized thoroughly to ascertain the system's dependable performance in various scenarios.

7.2 Test Plan

The test plan outlines the method we use to test the ShikshaSetu system. Among other things, it details the testing approach, what tools we used, and the order of our testing activities. Our testing process was based on a well-organized plan that aimed at a thorough coverage of the entire system.

1. Testing Strategy

The following testing strategies were adopted:

- Unit Testing Each module was tested individually to verify its functionality.
- Integration Testing Combined modules were tested to check the proper working of their interaction.
- System Testing The entire system was tested as a whole.
- User Acceptance Testing (UAT) The system was tested by users to validate its usability and quality.

2. Test Environment

The system was tested in the following environment:

- Operating System: Windows
- Browser: Google Chrome
- Backend Server: Flask
- Database: PostgreSQL
- Deployment Platforms: Vercel and Render

7.3 Test Result

The system seems to be performing well, as confirmed by the test. Tests were conducted both at the functional level as well as at the system level.

Test Case	Expected Output	Result	Status
User Registration	Account created & stored in DB	Successful	Pass
Store user data	Data stored correctly	Data stored correctly	Pass
Fetch data	Data retrieved	Data retrieved	Pass
Send request	Proper response	Proper response	Pass
Load pages	Pages displayed correctly	Pages displayed correctly	Pass

Table 7.1 Test Results Summary

7.4 Insights

Through assessing the ShikshaSetu platform, we managed to gain a deeper understanding of how well it works and how easy it is for people to use it.

- The system offers a very smooth and easy-to-use interface, which helps users to quickly get used to it and locate what they need effortlessly.
- The parts of the system that the users see and interact with, the server side, and the database are well connected, very efficiently working with each other and are dependable.
- In a usual scenario, the time it takes for the system to reply is more than acceptable.
- No matter what type of device is used, the system can cater to the users needs very well and make sure that they are able to use the system.
- Besides that, some slight fine-tuning can be done in system performance and scalability aspects.

Generally speaking, the system is a great performer and it is aligned perfectly with what it aims to deliver [8].

CHAPTER 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

8.1 Social Aspects

The ShikshaSetu system significantly contributes to the social development of people by increasing accessibility, inclusiveness, and the learning experience as a whole. Nowadays, with digitization of numerous aspects of life, educational resource awareness is no longer tied down to a physical classroom or library only. The proposed system effectively closes the gap between users and education content by offering a unified digital platform that is remotely accessible 24/7.

One of the greatest societal advantages of the system is how it can do away with the factor of location as a hurdle. Pupils and other individuals in under-equipped regions often lack access to quality educational resources. When learning content is posted online, the system acts as a safeguard for these users not being excluded from chances of improving their knowledge. This measure assists in narrowing the digital divide and upholds the ideal of education being equally available to all.

The system also boosts user engagement by giving a very easy and clear interface. Users will be able to move around the platform's different sections without any problem even if they are not technically very skilled. This is a very good way of getting people from different ages and different backgrounds to start using digital platforms for educational purposes. Besides that, the system also helps promote self-learning and independent study.

Users are able to learn anytime they want, they can also go through the content as many times as they wish, and they're able to learn according to their own schedule. Such a degree of freedom is indeed very helpful, especially for students who are preparing for their exams or for professionals who are planning to acquire new skills.

Besides, the system aligns very well with the idea of digital transformation in education. With the gradual shift of institutions to online platforms, systems like ShikshaSetu greatly assist in designing a well-ordered and planned environment for handling educational content. Besides that, it is a platform that fosters interaction and the sharing of knowledge among the users.

Besides, the platform also helps in some ways to lessen reliance on tangible infrastructures like libraries and printed works. Apart from saving time and energy this also leads to a more efficient and contemporary learning method. Ultimately, the ShikshaSetu platform system is a good force in society as it improves easy availability, encourages inclusiveness and facilitates digital learning programs.

8.2 Legal Aspects

The ShikshaSetu system's design and installation have been accompanied by a number of legal issues that need to be resolved through the harmonious integration of the data protection legislation, intellectual property rights, and system security standards. Thoroughly dealing with these legal issues is very important for the system to be a trustworthy, credible, and accountable one. The protection of user data, which is one of the main legal issues in any digital system, is foremost among the concerns in ShikshaSetu.

Since the ShikshaSetu platform is used for interaction with the users with their personal information and other data being the main focus, the implementation of the data protection principles demands that the basis of confidentiality, integrity, and availability of the data are basically maintained. Besides ensuring data protection, ShikshaSetu also focuses on data privacy through adherence to privacy rules.

The platform takes care that the personal data of the users are collected only for the purposes authorized by the users and that their data is not used for other purposes or disclosed to third parties without their consent.

We have also considered intellectual property rights while developing the system. Any educational content, materials, or resources used within the platform are managed in a way to avoid copyright infringement. The system only allows uploading and sharing of authorized content with users.

Cloud platforms deployment introduces additional legal considerations regarding hosting and data storage. Besides, the system uses third-party services like Vercel and Render that comply with standard security and compliance protocols. It is equally important to verify that these platforms comply with the laws and provide safe environments for hosting the applications.

On top of that, the system has to comply with the software licensing regulations in general. The open-source technologies that are a part of the development process and are allowed for use have been licensed to enable their modification and distribution without any legal conflicts.

Finally, ShikshaSetu system has been developed taking into account legal requirements, user data protection has been prioritized, rights of intellectual property have been respected, and all the operations have been performed within the legal boundaries.

8.3 Ethical Aspects

Incorporating ethical considerations is an indispensable part of system development because through these considerations we ensure that the technology is not only targeted towards responsible use but it also benefits the society.

ShikshaSetu indeed gears its features towards compliance with ethical principles such as fairness transparency accountability, and user privacy. One major ethical feature of the system is the safeguarding of user privacy. The system is designed in a way that personal information can neither be misused nor accessed without permission. In addition to that, users always have a secure space where their data is being responsibly managed [6].

Only ethical data collection measures are carried out whereby information is exclusively used for its purpose. Besides that, the system advocates equality by giving all users unhampered access. It does not make any biased judgments based on users' factors like location, social background, or computer skills. Moreover, the software is intended to be a community one with a wide range of users deriving benefits from the features it offers.

Transparency is yet another significant ethical principle embodied in the system. The workings of the system are distinctly laid out in a manner that users are given truthful and complete information. There is no deceptive content or any secret operations that could potentially lead to mistrust among users.

The system, on the other hand, emphasizes that the users receive truthful and useful information. Significant steps are taken not only to keep the details accurate but also to prevent the spread of wrong or misleading information. This is very crucial in a learning platform where the users depend on the system for their education.

Moreover, the system does not indulge in unethical acts like exploiting user behavior or secretly tracking user actions. It acknowledges the user's freedom and grants the users the liberty to use the platform without any hidden restrictions.

The system developers are accountable for upholding ethical principles at every stage of the project life cycle. That means regular checking, refreshing, and enhancing the system to make sure it is in line with the ethical norms.

In essence, the ShikshaSetu system observes ethical principles by safeguarding privacy, ensuring fairness, being transparent, and making a responsible use of technology.

8.4 Sustainability Aspects

Today, sustainability plays a major role in designing new systems because it centers on lowering one's carbon footprint and does so while still making the best use of available resources.

Digitally transforming the ShikshaSetu platform as well as reducing the dependency on physical materials are two ways, among others, through which the system aligns with sustainability. For instance, the biggest way the platform helps environment is by the major leverage of the reduction in paper and other printed materials.

Therefore, in the conventional education, the students have physical books, notes, and other documents, and for the most of time, it is their main dependency. In this regard, by offering digital resources to the users, the platform helps immensely in cutting down to paper usage and consequently is a helper to the nature.

Besides, usage of the cloud infrastructure effectively helps in achieving sustainability as well. Instead of depending on individual hardware systems, this platform works on cloud services, which are equipped to be energy efficient. That is the reason, that cloud-based platforms are capable of handling large operations with minimum wastage of resources and this characteristic of them is making them a more environmentally friendly option.

Moreover, the system is engineered to be highly scalable and flexible, i.e. it can majorly deliver without coming to shortcoming of resources. In other words, the function of scaling is that the

platform can still offer well even in the event when there is growth, and at the same time, the need for frequent hardware upgrades is drastically reduced.

Another angle of sustainability relates to the system's viability over a long period of time. ShikshaSetu is built on up-to-date technologies which enable it to be easily updated and maintained.

So, the system is capable of growing time after time without a major overhaul. Besides that, as the system is digital, it limits physical transportation since users directly get the resources online at their convenience rather than making a trip to workstations.

Such measures lead to a lowering of carbon emissions and at the same time, to support of green practices. Besides that, the product pushes users to go for digital learning which is a worldwide sustainable development goal. Therefore, the system, by lessening the environmental impact and promoting resourcefulness, embraces the sustainability concept in education.

8.5 Safety Aspects

Safety is one of the fundamental considerations in the design of any system, especially when it comes to the protection of user data, assurance of system reliability, and thwarting of unauthorized access.

ShikshaSetu system leverages a number of safety mechanisms to offer users a secure and dependable platform. Secure user authentication is one of the major safety features of the platform. The system restricts access to accounts only to authorized users by using effective login procedures. Passwords and other confidential data are treated with utmost security to avoid any leak of information.

Besides the above, data validation is one of the crucial safety steps. Through the validation of all user entries, the system not only avoids any errors but also limits the ways in which attackers may use the system to their advantage (e.g. through injection attacks). The outcome is a system that continues to function well and securely.

The system has plans to keep the data untampered. Correctly managing the database is part of the plan to have data stored reliably and at the same time consistent. Methods of backup can be arranged to avoid data loss due to system breakdowns [5].

A deployment on trustable cloud platforms in turn is a step to system safety since they offer protection against downtime and technical failures. Cloud services come with features like load balancing, monitoring, and automatic recovery that help to maintain the system stability.

Checking the system status regularly and doing maintenance extensively will result in finding the possible problems and fixing them before the users get affected. With this method, the system will always stay secure and working without any problems.

Besides that, the system also guarantees the users that their data is kept safe and not used without their permission. The security arrangement gets regular updates to be able to hold up against the newly uncovered threats and vulnerabilities.

To sum it up, the ShikshaSetu system makes the safety one of its top priorities by coming up with an authentication that is secure, validation of data, trusted deployment, and double-checking all the time. That makes it a safe and reliable place for users.

CHAPTER 9

CONCLUSION

The ShikshaSetu system is a total and powerful solution to the problems of people struggling to get and use educational resources in a digital environment. The main aim of the project was to create a single platform through which users can get academic content easily, quickly, and reliably. This system went live and through it the project made a point in how modern internet tools can help learning and resource management be taken to the next level.

One big thing that the system achieved is making a simple and natural interface so individuals can work with the platform without running into any problems. The looks and feel part is done so well that the platform is easy to get around and can even be used on various gadgets like desktops, laptops, and phones. This makes the system very handy for everyday use since people may get the platform in different locations.

The system's backend built with Flask quite efficiently manages user requests, data processing, and working in harmony with other parts. The usage of a strong database with PostgreSQL also supports the efficient storage, retrieval, and management of data, hence the system stays trustworthy and for good use support multiple users at the same time.

Another major point of the project is that it integrates all system elements altogether very well. The frontend, backend, and database are linked to each other in such a way that there is smooth data flow and consistent performance.

This integration really shows how important a well-structured system design and proper implementation strategy are. When the system was evaluated and tested, it was found to be satisfying the expected functional and performance parameters. All the main blocks of the system were tested in different scenarios, and the system results after the tests showed that it is performing efficiently and reliably.

The system is quick in its response, processes data accurately, and performs stably, meaning it is fit for real-life implementations. Besides this, the project serves as a reminder that the social legal ethical, sustainability, and safety factors should be taken into account during system development. Considering these points, the system not only ensures the responsible use of technology but also makes a desirable contribution to society. Digitization through digital

solutions is a way to minimize reliance on natural resources and at the same time continue with the eco-friendly development support [7].

Still, the system does have a few weaknesses that could be rectified in the upcoming changes. For instance, system scalability improvement, implementation of additional, sophisticated features, and performance optimization during peak user loads are some of the aspects that can be looked upon.

Also, through the incorporation of cutting-edge technologies like ai, customized suggestion and better analytics, the user experience can be taken to a whole new level. In summary, ShikshaSetu has been able to achieve its goal of offering a platform for educational resource access that is dependable, effective and easy to navigate.

This project not only shows how technologies of today can be used to address issues of the world but also paves the way for further advancement and enlargement [1].

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BASE PAPER:

Kapp, K. M., The Gamification of Learning and Instruction: Game-Based Methods and Strategies for Training and Education, Wiley, 2012.

APPENDIX

I. Software Stack / Dependencies

• Backend Framework: Flask (Python 3.x).

• Database: PostgreSQL (game_db) for users & teachers, SQLite (translations.db) for translation cache.

• ORM & Extensions: SQLAlchemy, Flask-Babel, Flask-CORS, Flask-Mail.

• Other Libraries: requests, hashlib, threading (Lock), json, base64, io.BytesIO for API calls, caching, and translation handling.

II. Publications

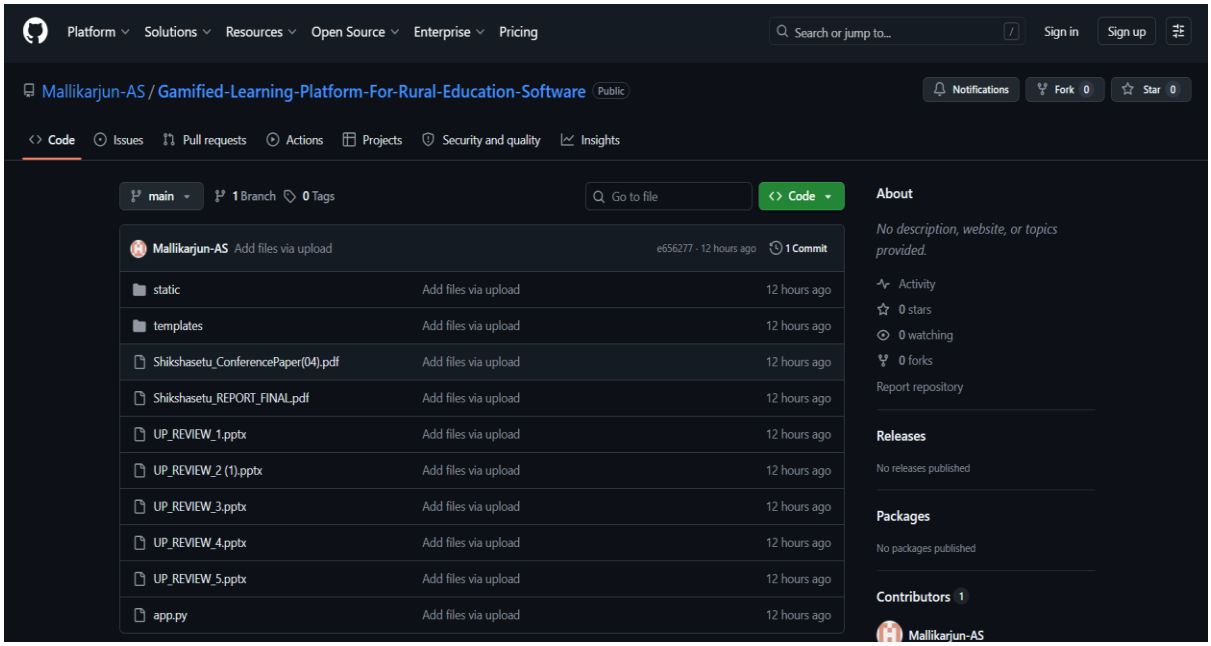
• Acceptance letter for conference/journal paper: Not applicable / To be added.

• Scopus URL: N/A.

III. Project Demo

• Source Code (Core Backend): app.py (Flask server with routes for login, registration, Student Dashboard, Games, translation, Teacher Dashboard, Leaderboard, and Report sent through email).

• GitHub Repository: https://github.com/Mallikarjun-AS/Gamified-Learning-Platform-For-Rural-Education-Software.git



IV. Few images of Project

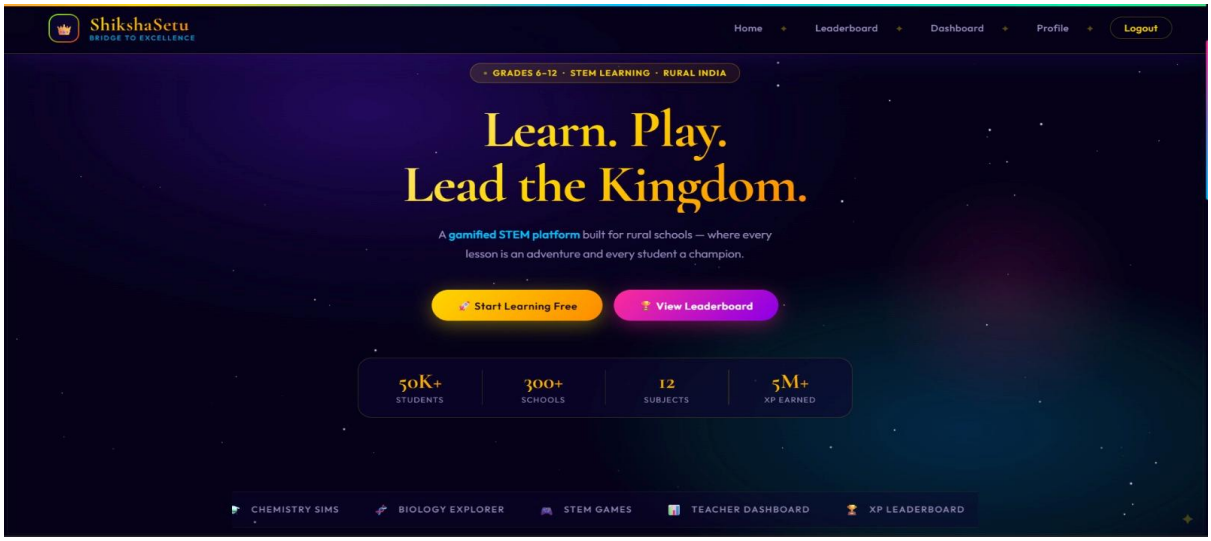


Fig A. Home Page

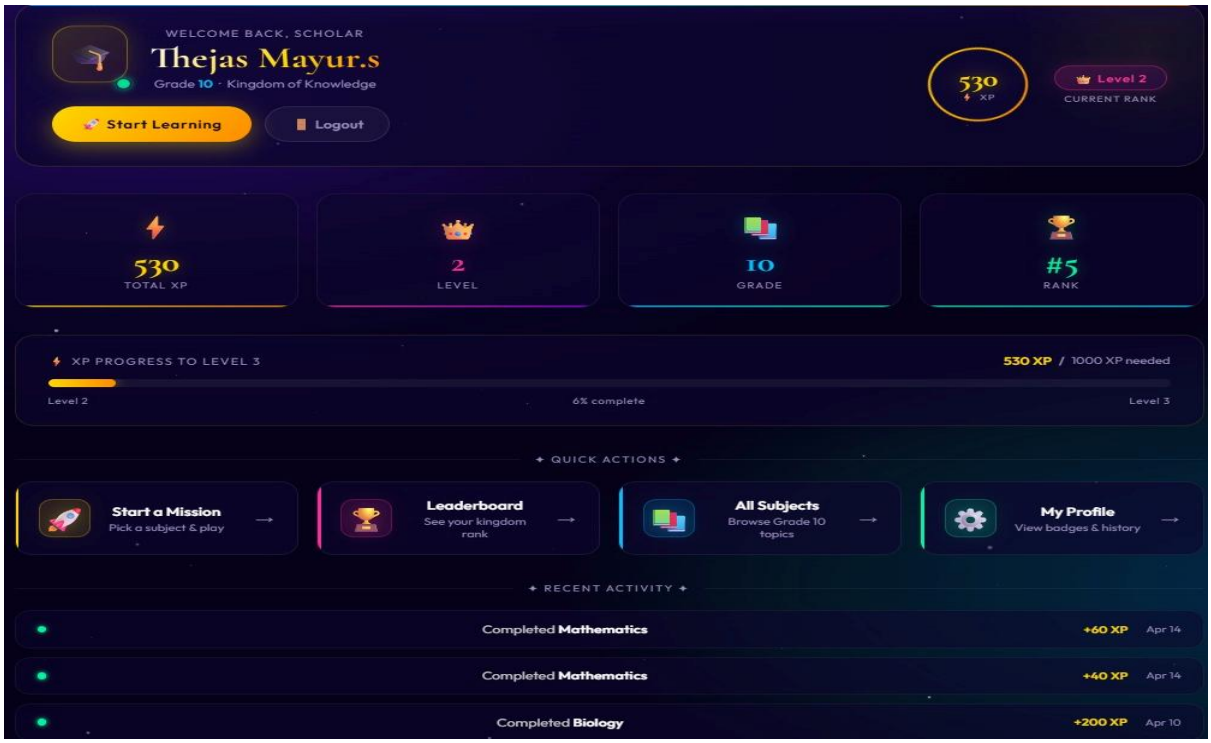


Fig B. Student Dashboard

Fig C. Leader Board

Fig D. Game 1

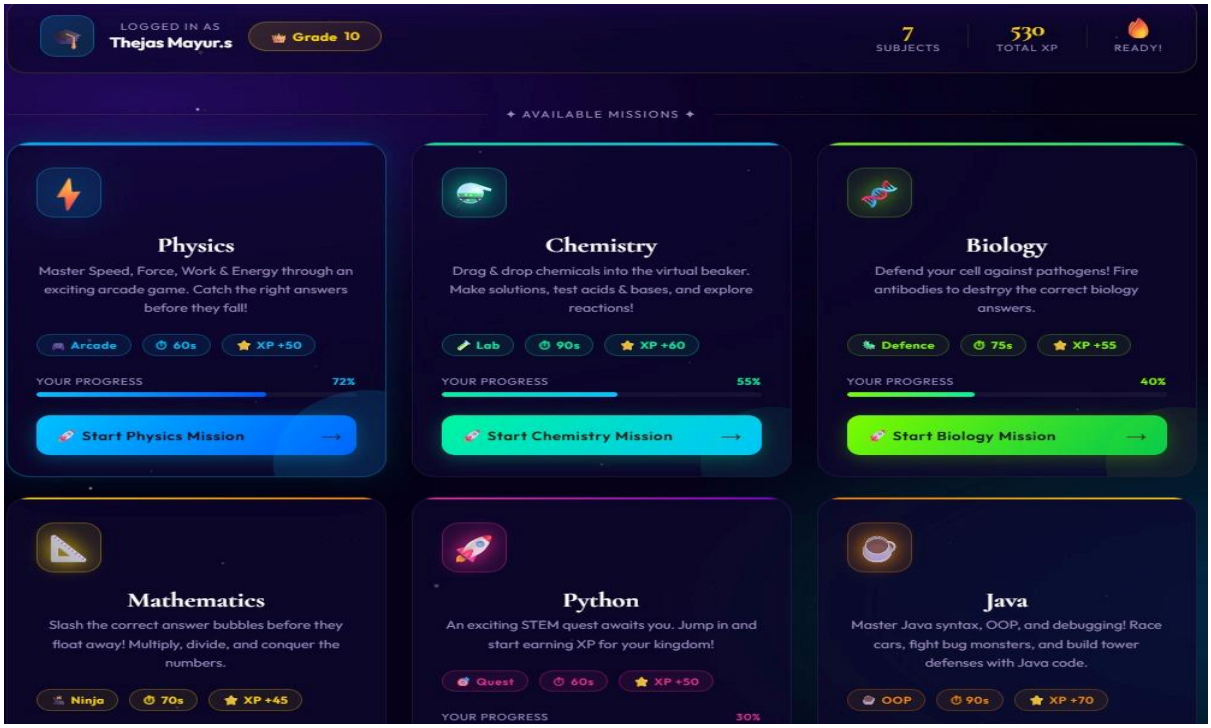


Fig E. Game Page

Fig F. Game 2

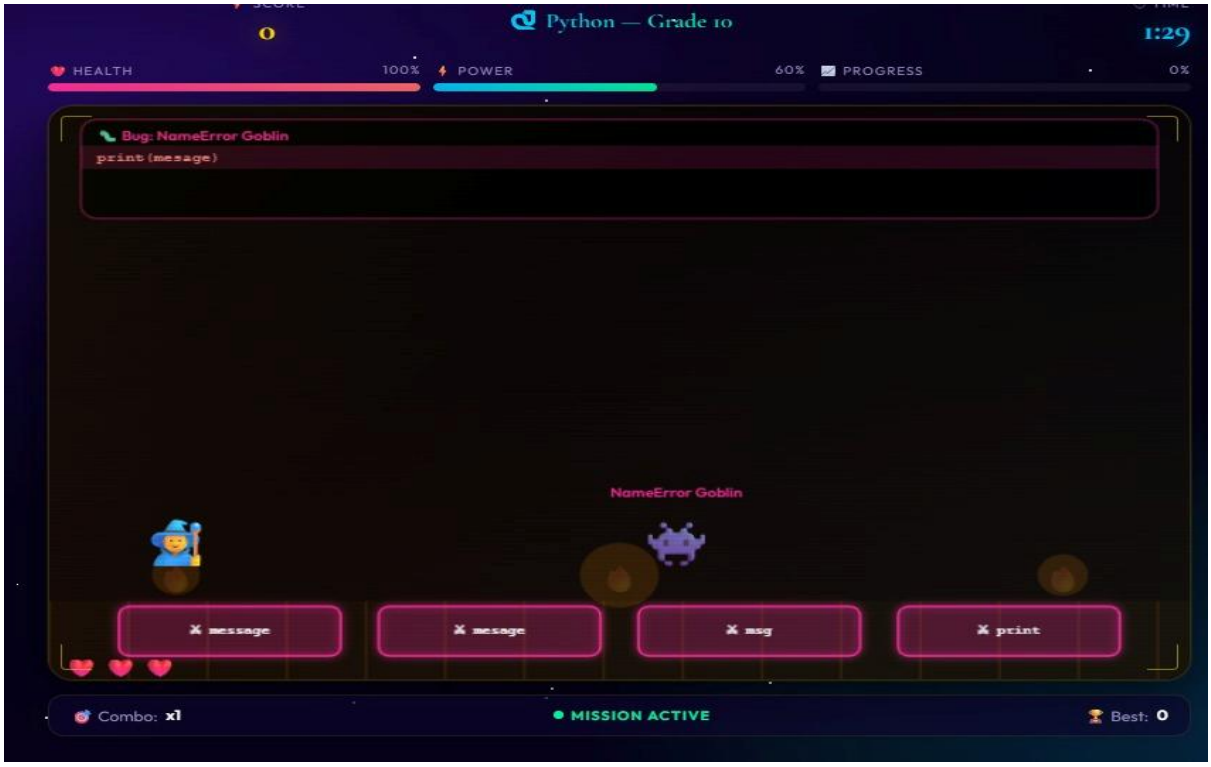

0
Python — Grade 10
1:29
HEALTH
100%
POWER
60%
PROGRESS
0%
Bug: NameError Goblin
print(mesage)
NameError Goblin
✗ message
✗ mesage
✗ msg
✗ print
Combo: x1
MISSION ACTIVE
Best: 0

Fig G. Game 3

TEACHER COMMAND CENTRE
Demo Teacher
STEM Faculty · ShikshaSetu Academy
8
STUDENTS
985
AVG XP
2
TODAY
Logout
Students with parent email: 0
Reports sent today: 0
8
TOTAL STUDENTS
7880
TOTAL XP EARNED
985
AVERAGE XP
5
HIGHEST LEVEL
7
ACTIVE LEARNERS
XP BY GRADE
1990
1570
1870
2450
Gr.10
Gr.6
Gr.7
Gr.8
LEVEL DISTRIBUTION
8
students
Lv1 (3)
Lv2 (2)
Lv3 (1)
Lv4 (1)
Lv5 (1)

Fig H. Teacher Dashboard

SHIKSHASETU COMPREHENSIVE PROGRESS REPORT

Generated on: 09 April 2026 at 03:10 PM

STUDENT INFORMATION

Student Name:	abcc
Email Address:	abc@gmail.com
Parent/Guardian Email:	abc@gmail.com
Current Grade:	Grade 6
Current Level:	Level 4
Total Experience Points:	1,735 XP
Member Since:	06 April 2026
Global Ranking:	#3 of 5 students
Performance Percentile:	Top 60.0%

PERFORMANCE METRICS

Total Games Played	2	**Total XP Earned**	1,735
Highest Score	1280	**Average Score**	867
XP Growth Rate	+-64.5%	**Progress to Next Level**	47%
Current Streak	1 days	**Longest Streak**	1 days

COMPLETE SUBJECT PERFORMANCE

Subject	Games	XP Earned	Best Score	Accuracy
Physics	1	1,280	1280	Not yet
Chemistry	0	0	0	Not yet
Biology	1	455	455	Not yet
Mathematics	0	0	0	Not yet
Python	0	0	0	Not yet

TIME-BASED PROGRESS

Period	Games Played	XP Earned	Average Score	Status
Last 7 Days	2	1,735	867	Active

Fig I. Report Sent Through Mail

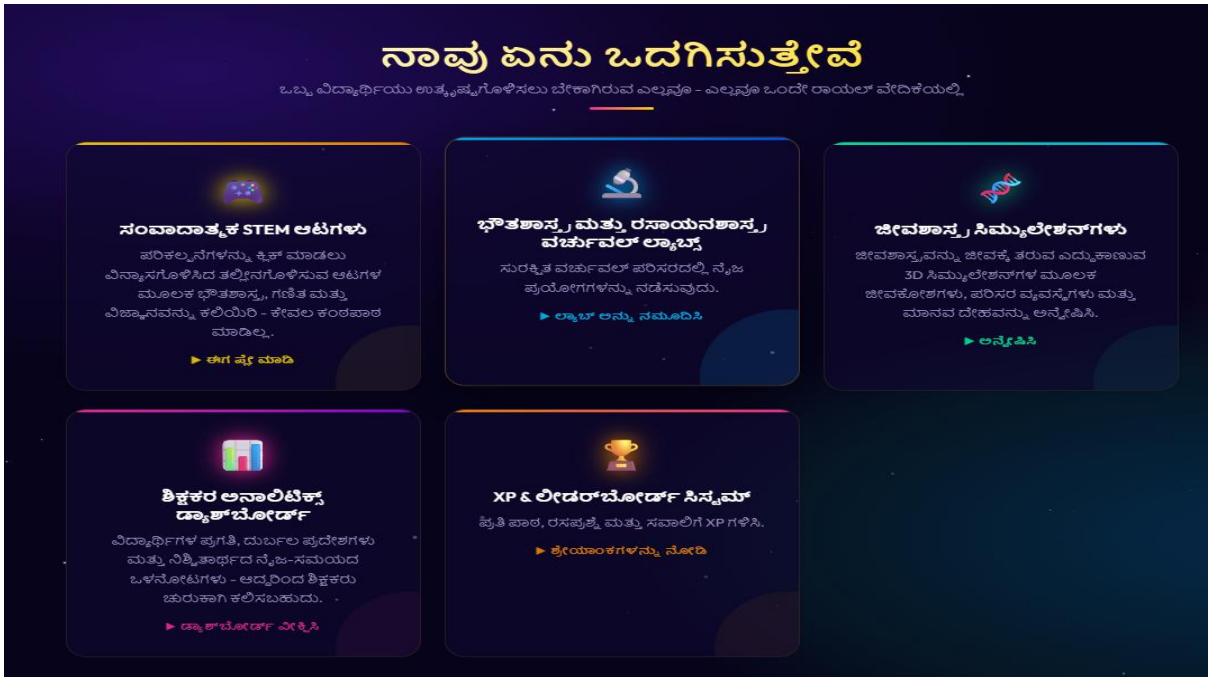


Fig J. Multilingual [Kannada]

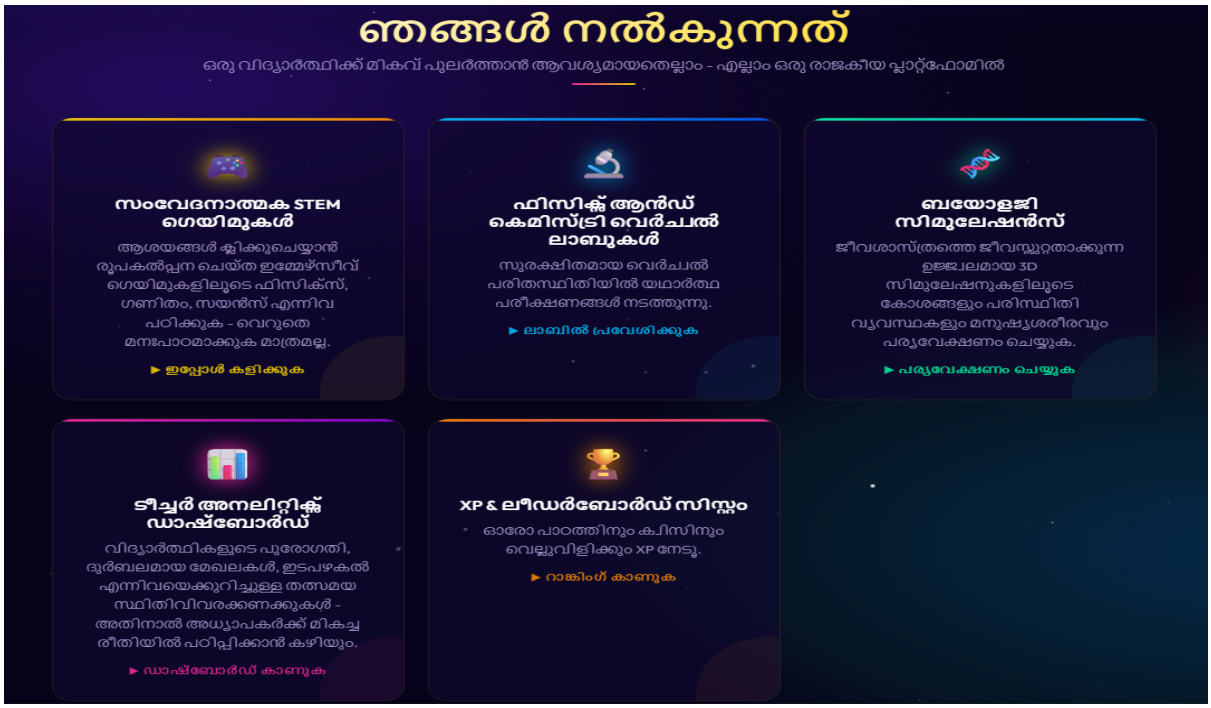


Fig K. Multilingual [Malayalam]